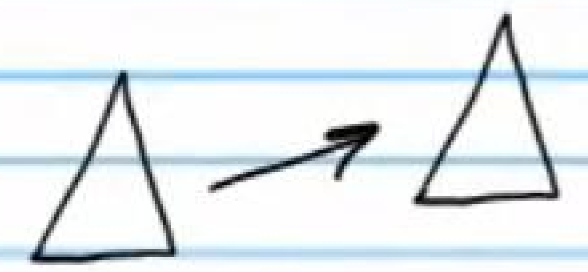
Name
Date
Course

Transformations and the Library of Relations (Part III)

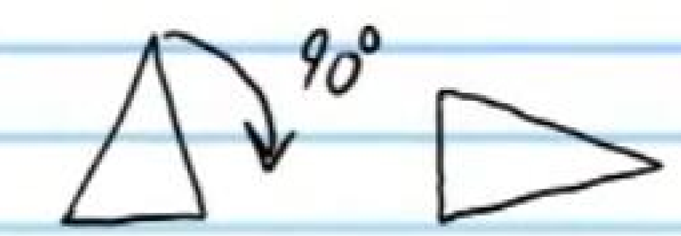
Review on Transformations

I) Translation



II) Rotation.

★ There must be a center of rotation



No Change in Size.

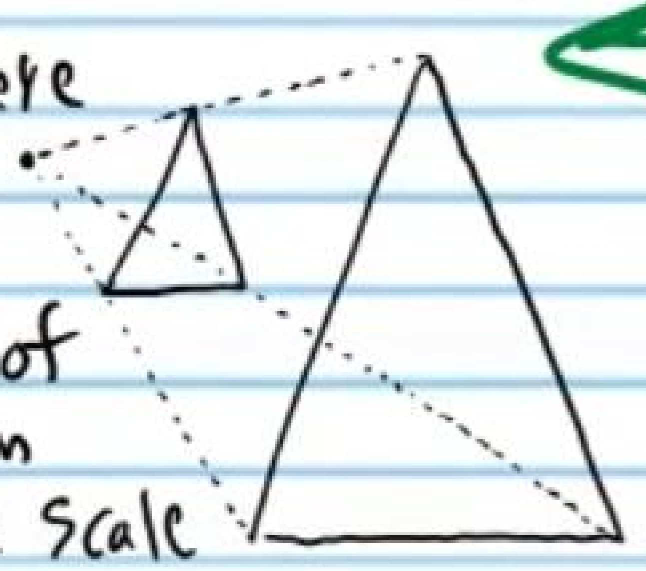
$\boxed{\text{Rigid}}$

III) Reflection

★ There must be a line of reflection.

IV) Dilation.

★ There must be a center of dilation and a scale factor.



Size Changes.

$\boxed{\text{Non-Rigid}}$

Relations

Relation: A connection between two values (usually called X and Y).

Three Ways to Represent a Relation

Examples

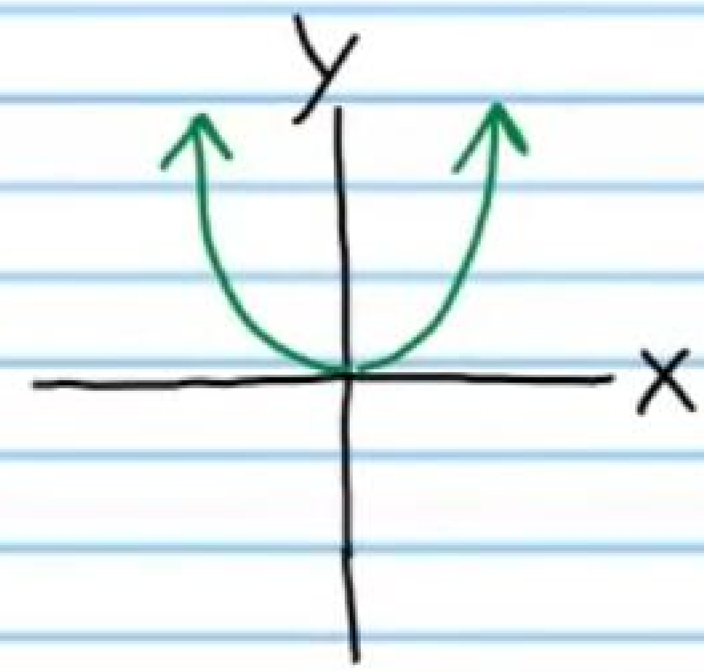
I. Equation $y = x^2$ $y = 3^x$

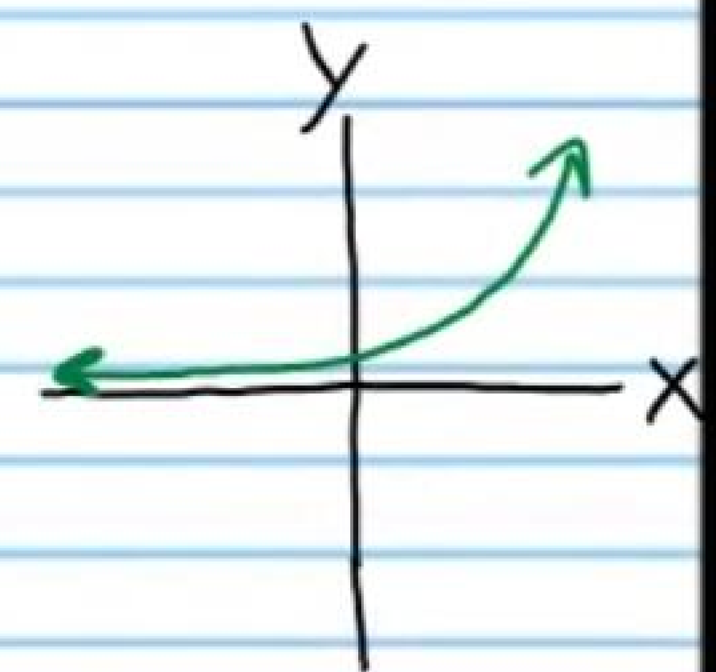
II. Table

X	-2	-1	0	1	2
Y	4	1	0	1	4

X	-2	-1	0	1	2
Y	1/9	1/3	1	3	9

III. Graph

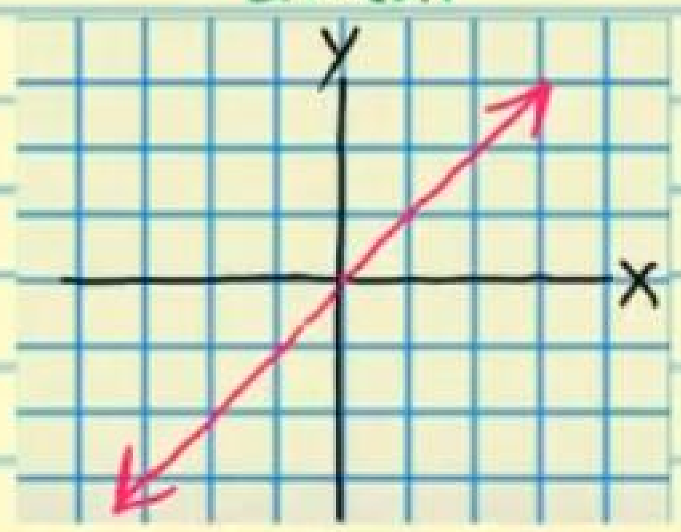




The Library of Relations

★ Also called "parent" relations.

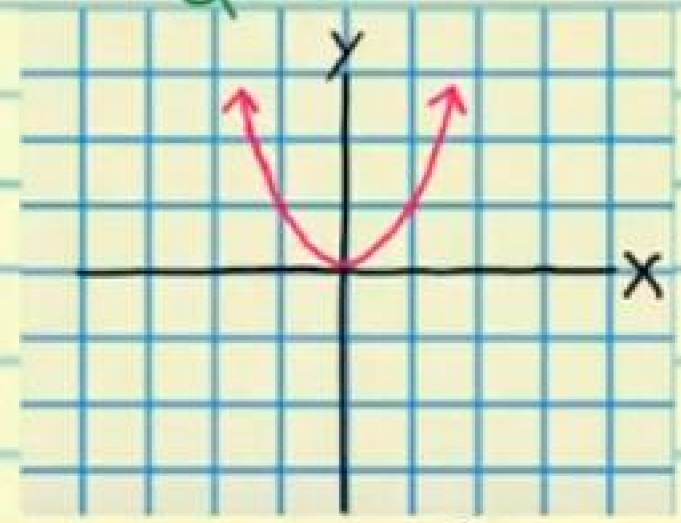
Linear



$y = x$

x	-2	-1	0	1	2
y	-2	-1	0	1	2

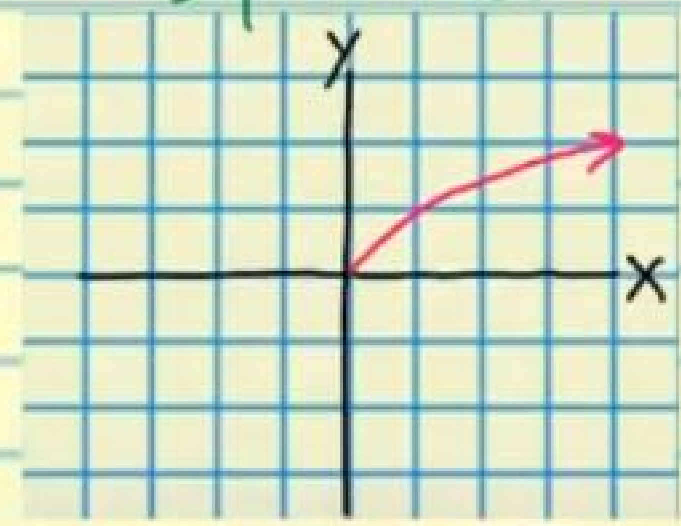
Quadratic



$y = x^2$

x	-2	-1	0	1	2
y	4	1	0	1	4

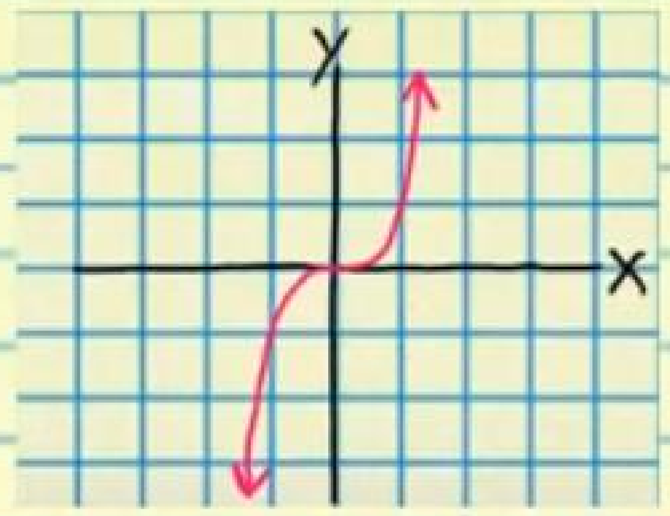
Square Root



$y = \sqrt{x}$

x	0	1	4	9	16
y	0	1	2	3	4

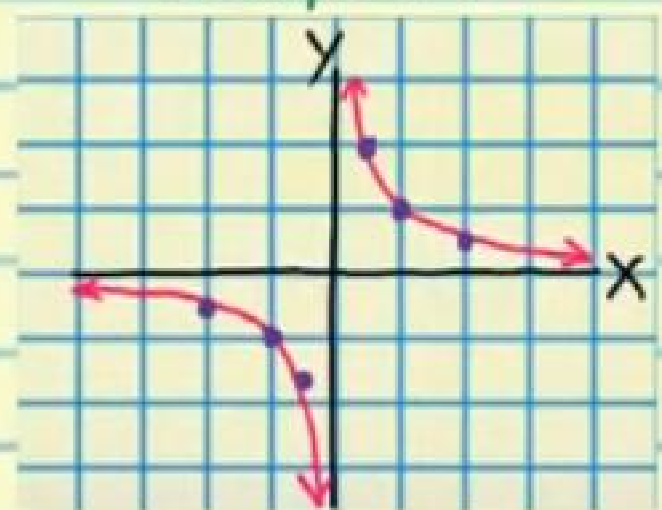
Cubic



$y = x^3$

x	-2	-1	0	1	2
y	-8	-1	0	1	8

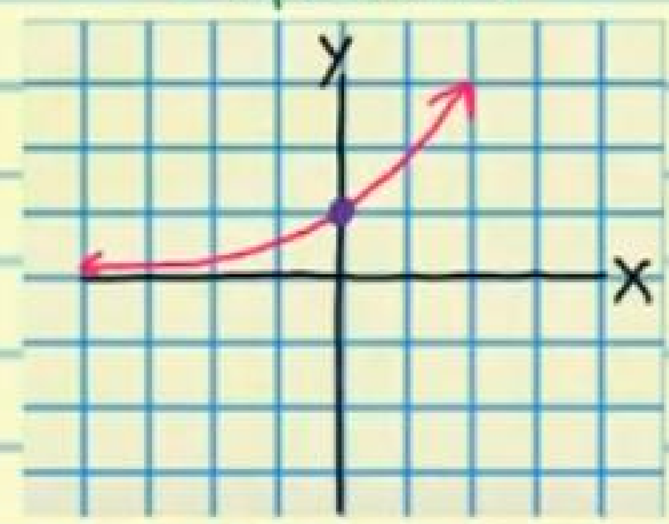
Reciprocal



$y = 1/x$

x	-2	-1	-½	0	½	1	2
y	-½	-1	-2	und	2	1	½

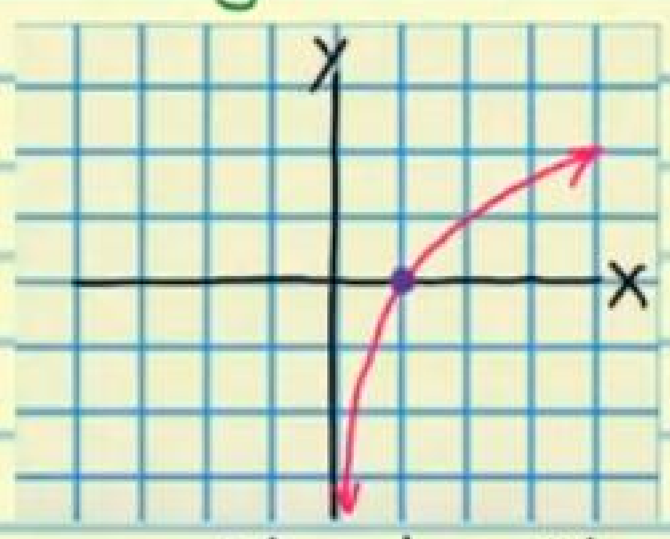
Exponential



$y = a^x,\ a > 1$

x	—	—	0	—	—
y	—	—	1	—	—

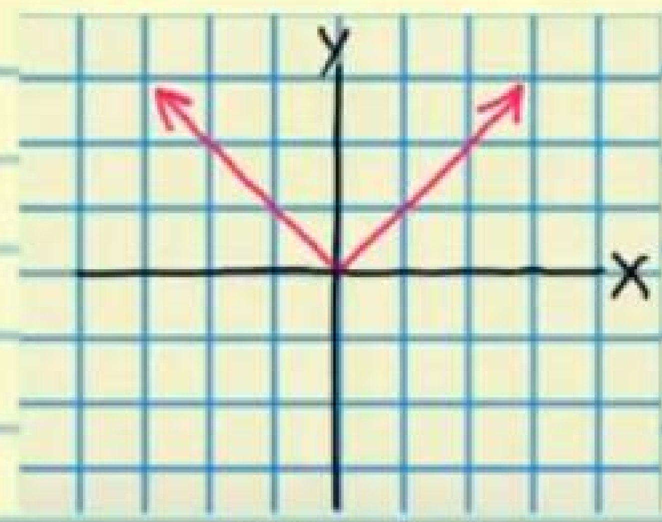
Logarithmic



$y = \log_a x$

x	0	1	—	—	—
y	und	0	—	—	—

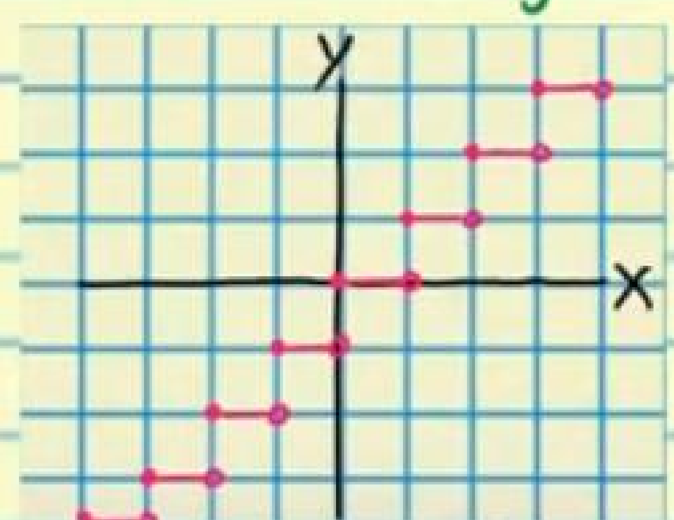
Absolute Value



$y = |x|$

x	-2	-1	0	1	2
y	2	1	0	1	2

Greatest Integer



$y = [x]$

x	-2⅓	-1	0	1/2	1½
y	-3	-1	0	0	1

The Transformation Properties

Translation

If y is replaced with y+k in an equation, then the graph of the new equation is the same as the graph of the original equation, but moved k units down (assuming that k is positive).

If y is replaced with y−k in an equation, then the graph of the new equation is the same as the graph of the original equation, but moved k units up (assuming that k is positive).

If x is replaced with x+k in an equation, then the graph of the new equation is the same as the graph of the original equation, but moved k units to the left (assuming that k is positive).

If x is replaced with x−k in an equation, then the graph of the new equation is the same as the graph of the original equation, but moved k units to the right (assuming that k is positive).

Reflection

If y is replaced with −y in an equation, then the graph of the new equation is the same as the graph of the original equation, but reflected across the x-axis.

If x is replaced with −x in an equation, then the graph of the new equation is the same as the graph of the original equation, but reflected across the y-axis.

Dilation

Vertical Compression

If y is replaced with ky in an equation, then the graph of the new equation is the same as the graph of the original equation, except every y-value is divided by k (assuming that k is positive).

Vertical Stretch

If y is replaced with y/k in an equation, then the graph of the new equation is the same as the graph of the original equation, except every y-value is multiplied by k (assuming that k is positive).

Horizontal Compression

If x is replaced with kx in an equation, then the graph of the new equation is the same as the graph of the original equation, except every x-value is divided by k (assuming that k is positive).

Horizontal Stretch

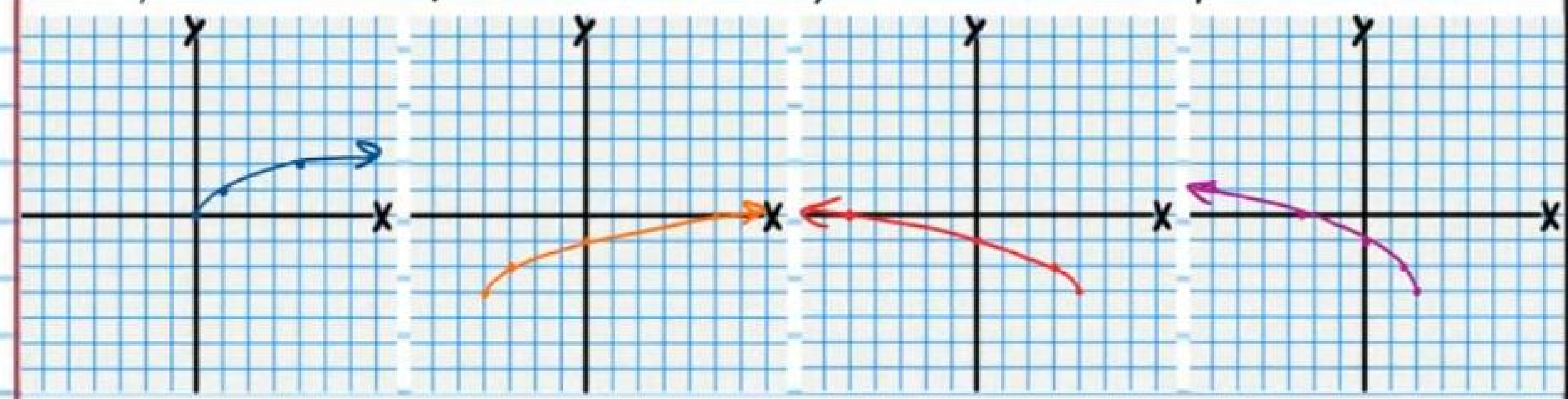
If x is replaced with x/k in an equation, then the graph of the new equation is the same as the graph of the original equation, except every x-value is multiplied by k (assuming that k is positive).

Graphing More Complicated Equations Using Transformations

Graph each equation below by performing multiple transformations on the parent graph. If a graph has asymptote(s), draw them as dotted lines.

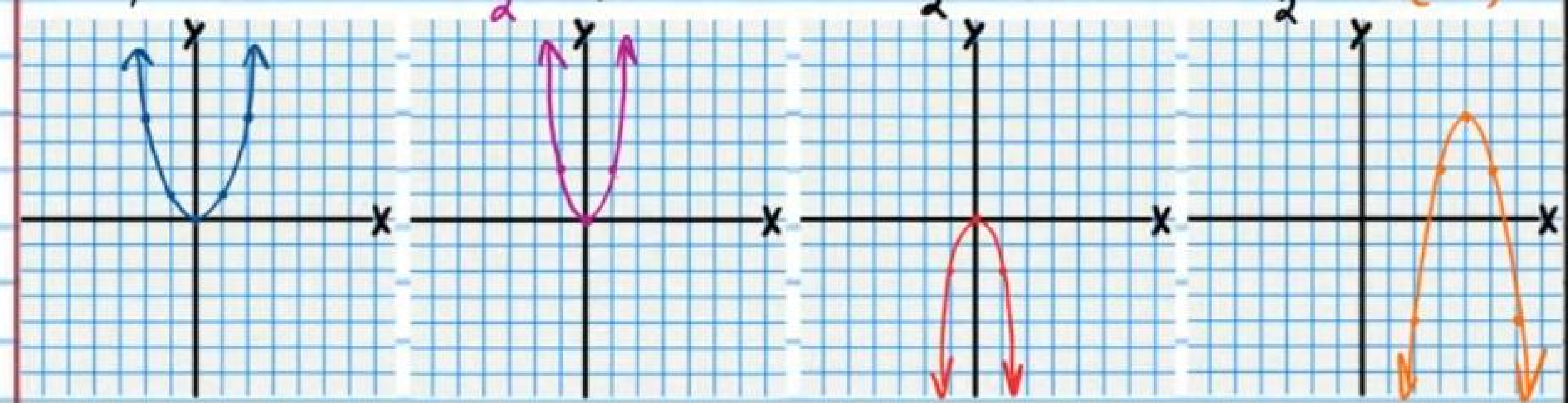
① $y+3=\sqrt{-2x+4}$

Parent Relation	Transformation #1	Transformation #2	Final Transformation
	Translation	Reflection	Dilation
$y=\sqrt{x}$	$y+3=\sqrt{x+4}$	$y+3=\sqrt{-x+4}$	$y+3=\sqrt{-2x+4}$



② $\frac{-(y-4)}{2}=(x-4)^2$

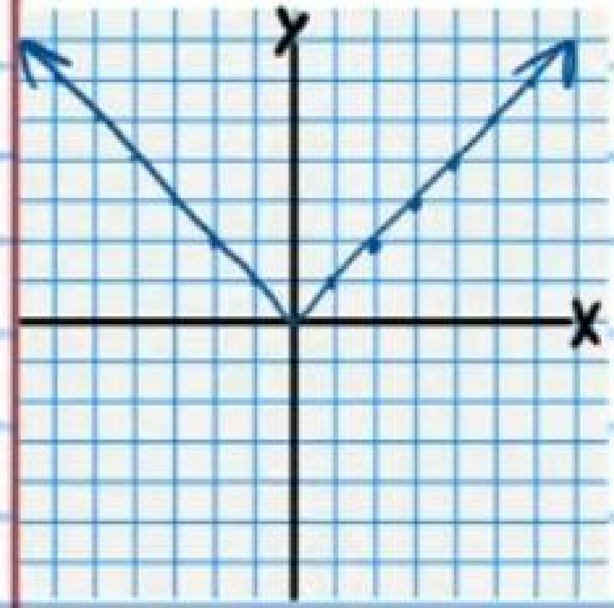
Parent Relation	Transformation #1	Transformation #2	Final Transformation
	Dilation	Reflection	Translation
$y=x^2$	$\frac{y}{2}=x^2$	$\frac{-y}{2}=x^2$	$\frac{-(y-4)}{2}=(x-4)^2$



③ $3y-1=|-(x+3)|$

Parent Relation

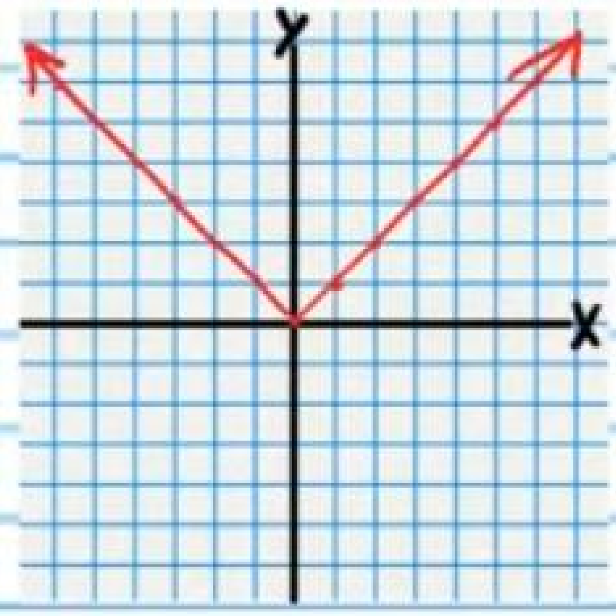
$y=|x|$



Transformation #1

Reflection

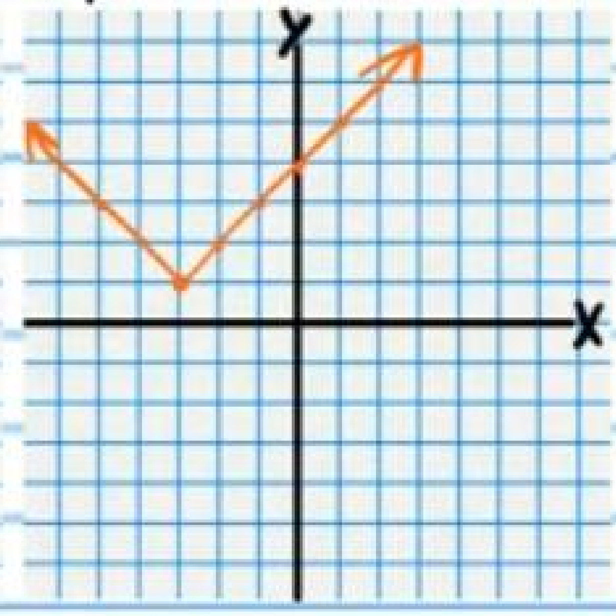
$y=|-x|$



Transformation #2

Translation

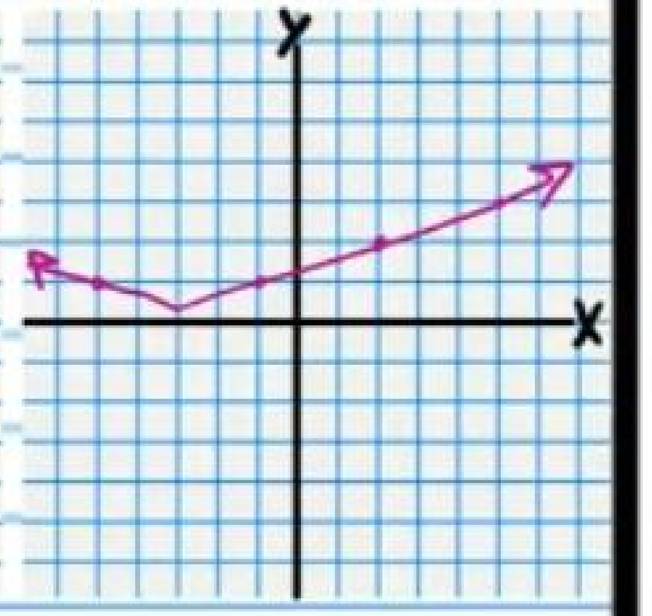
$y-1=|-(x+3)|$



Final Transformation

Dilation

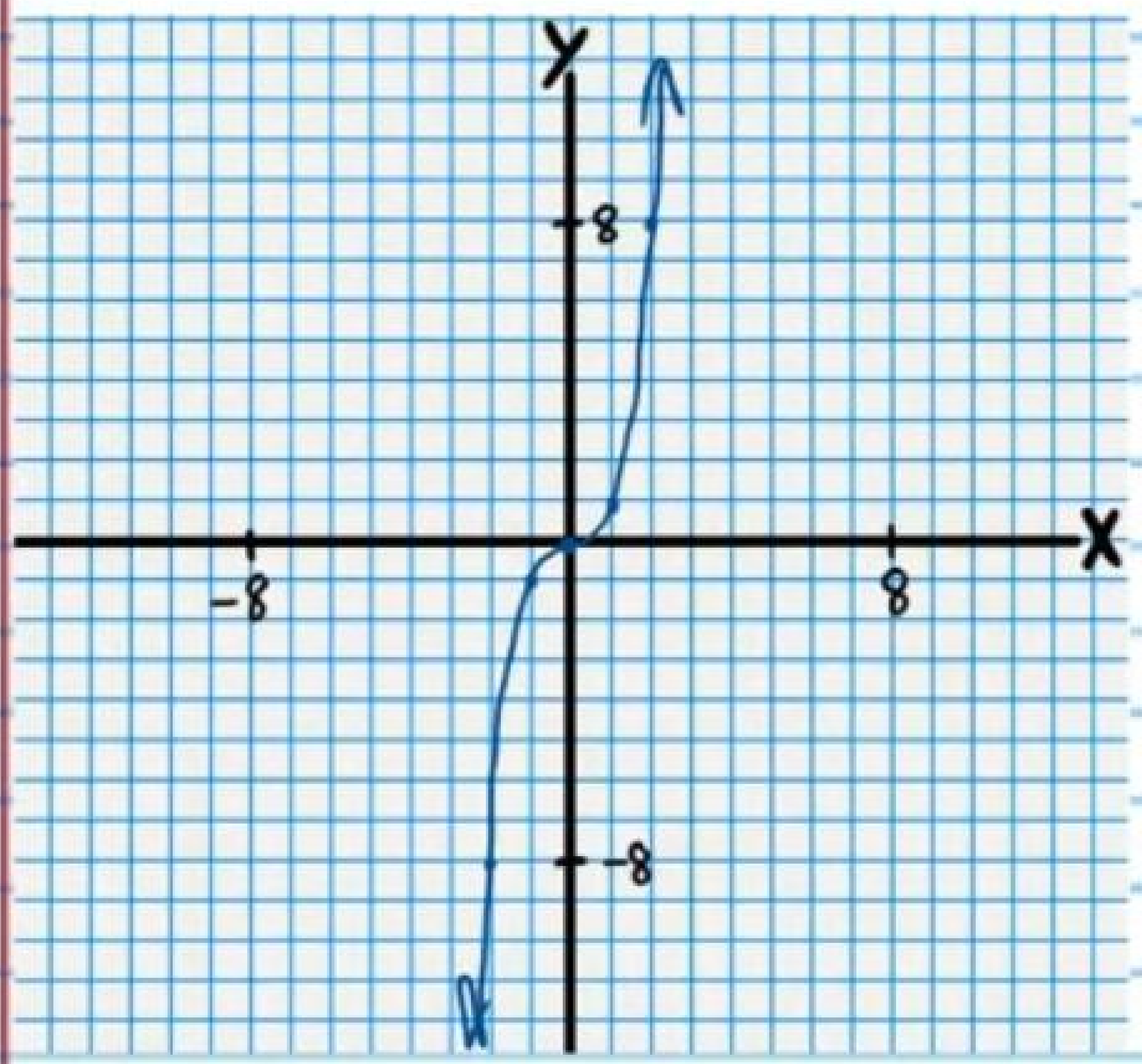
$3y-1=|-(x+3)|$



④ $-y+2=\left(\frac{x}{4}+1\right)^3$

Parent Relation

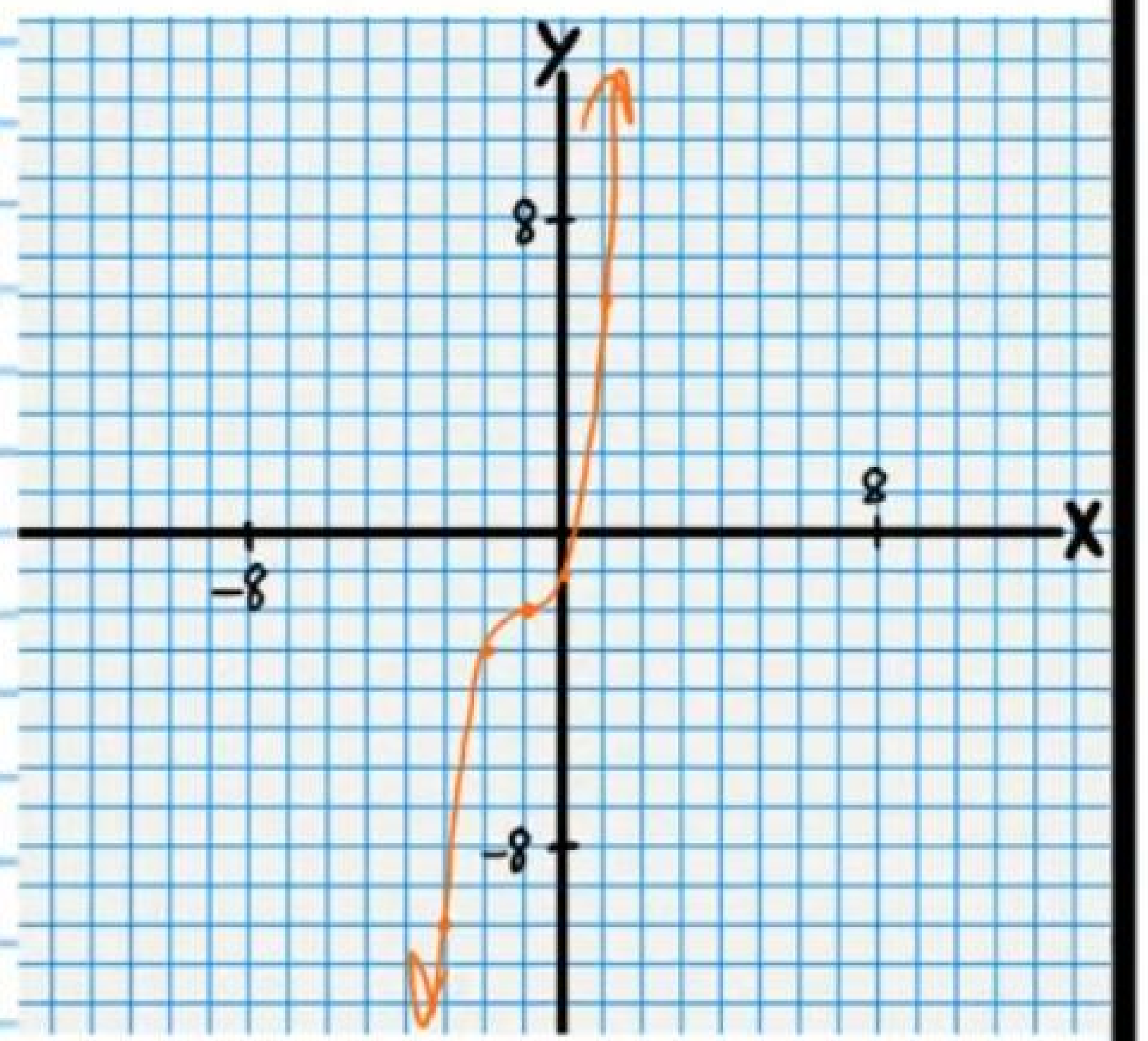
$y=x^3$



Transformation #1

Translation

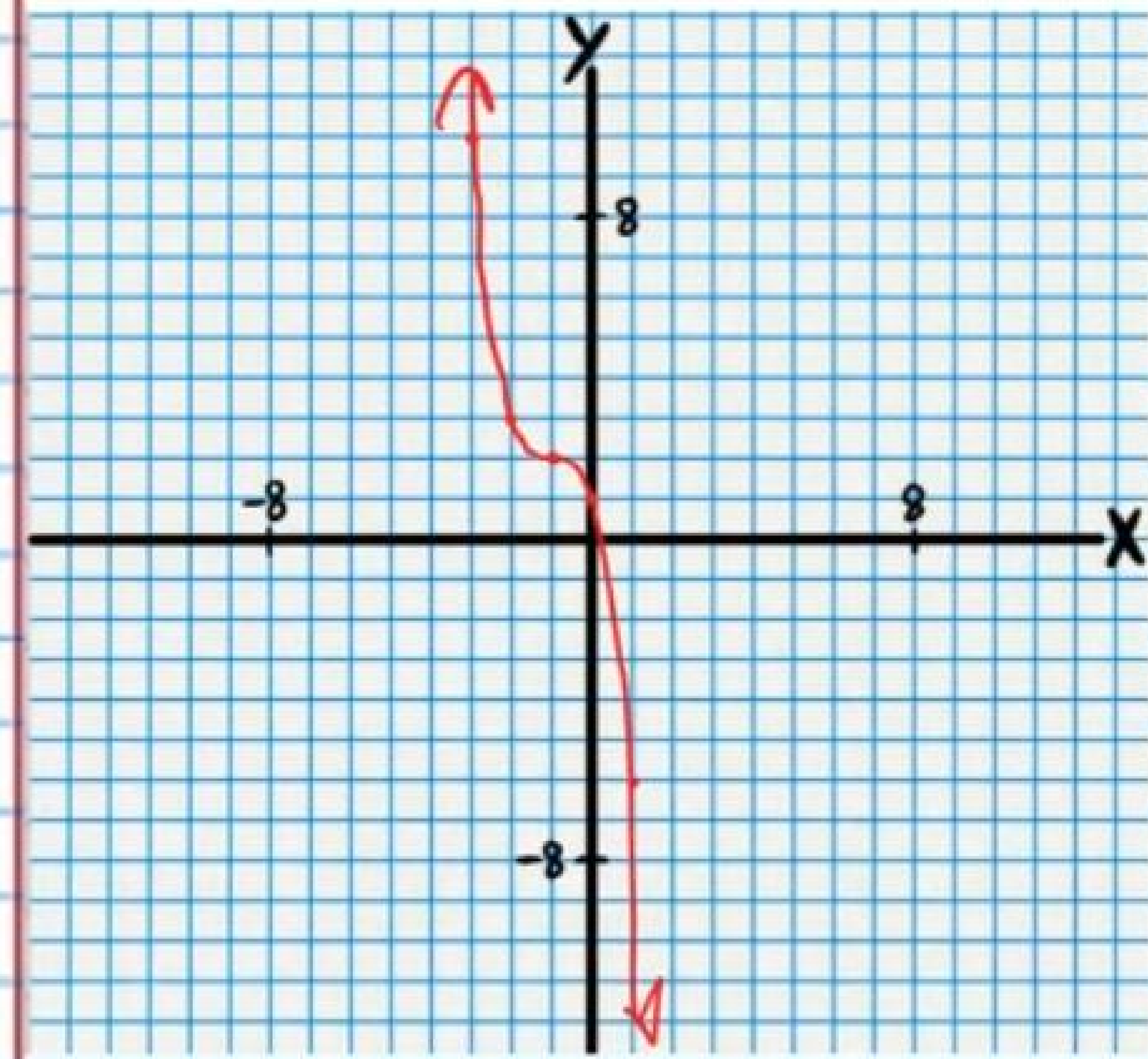
$y+2=(x+1)^3$



Transformation #2

Reflection

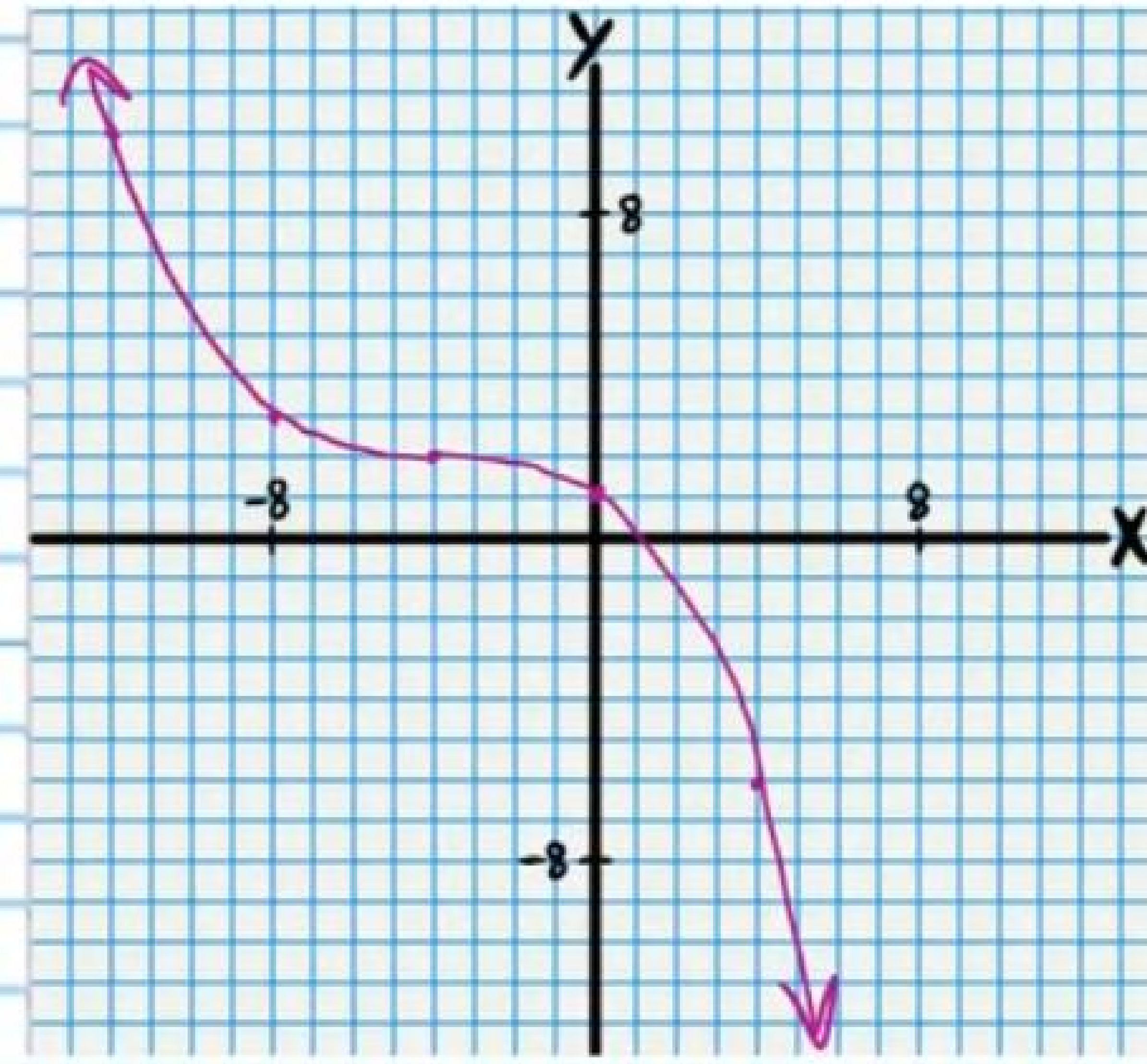
$-y+2=(x+1)^3$



Final Transformation

Dilation

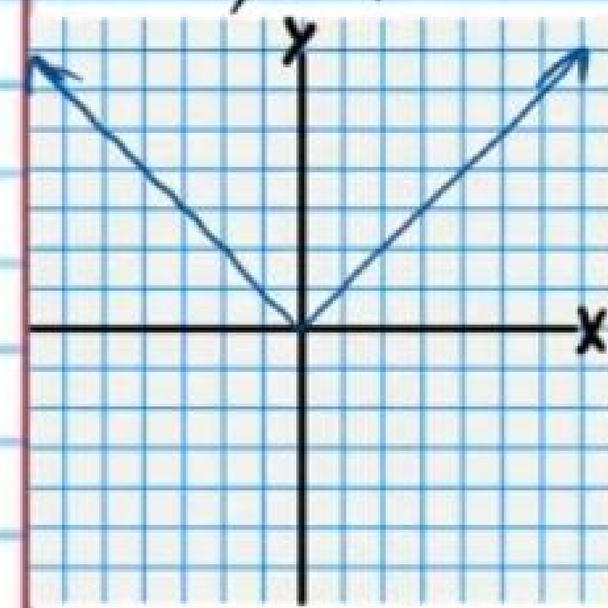
$-y+2=\left(\frac{x}{4}+1\right)^3$



⑤ $-(y-3)=|3(x-2)|$

Parent Relation

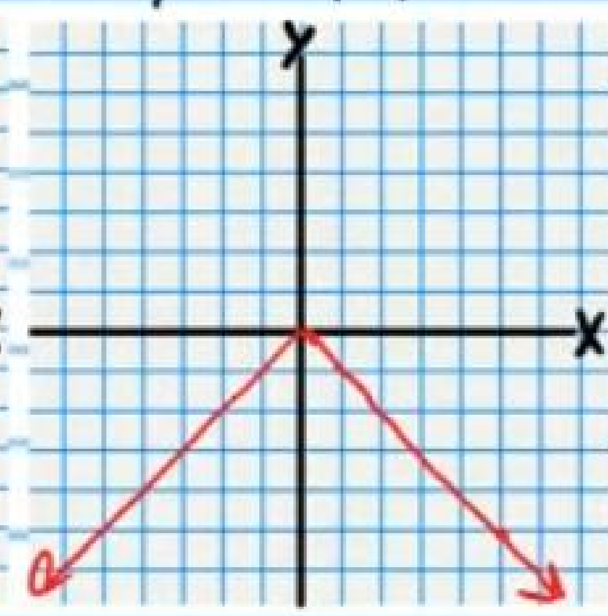
$y=|x|$



Transformation #1

Reflection

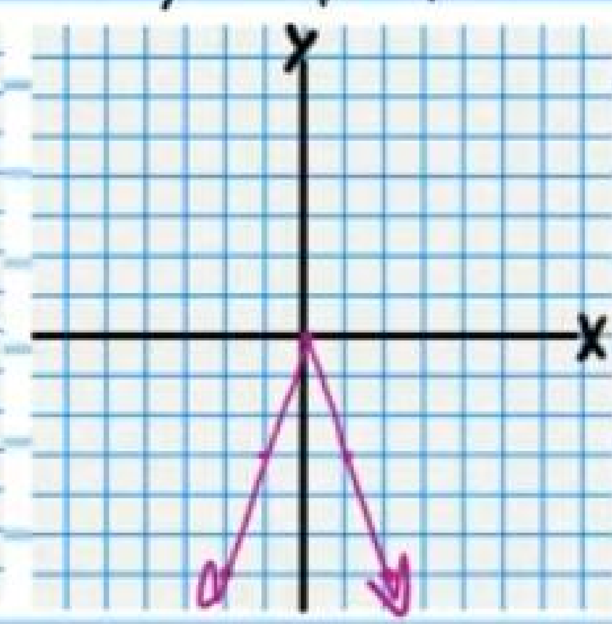
$-y=|x|$



Transformation #2

Dilation

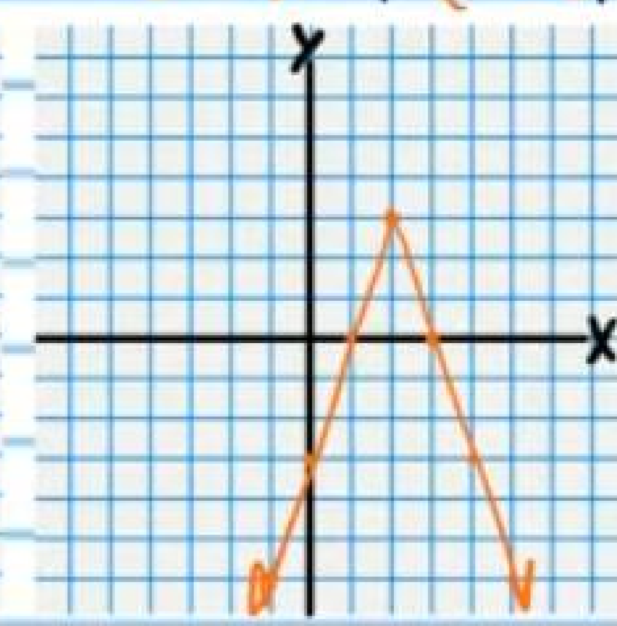
$-y=|3x|$



Final Transformation

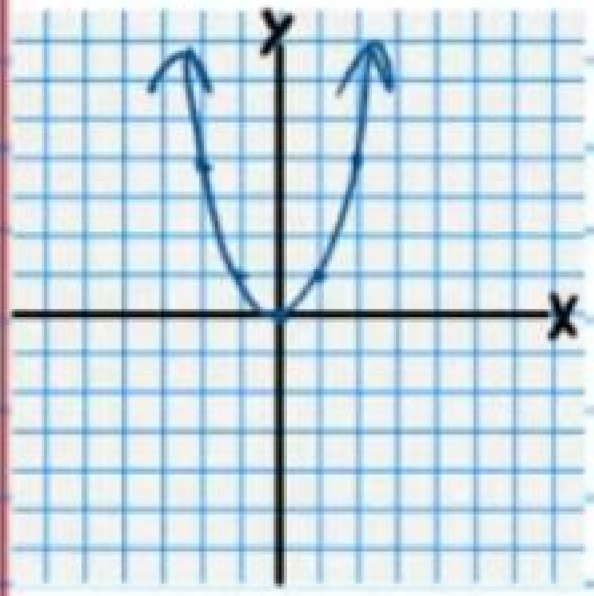
Translation

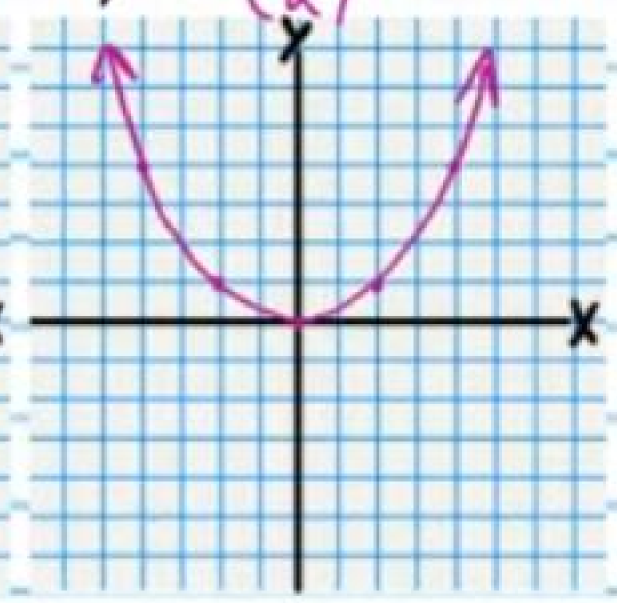
$-(y-3)=|3(x-2)|$

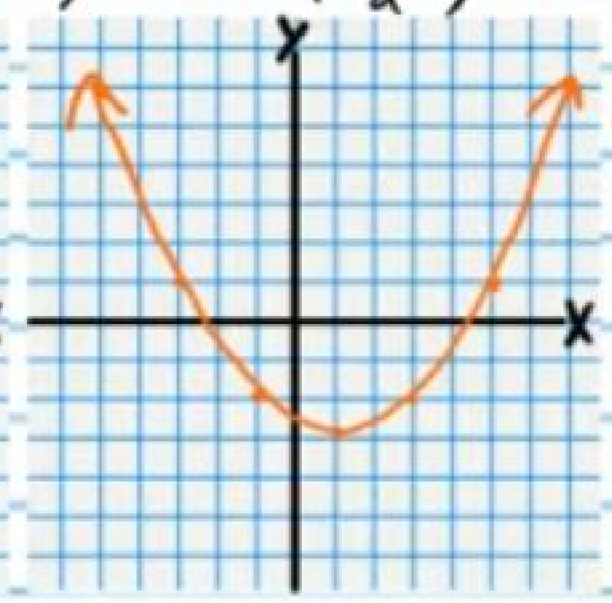


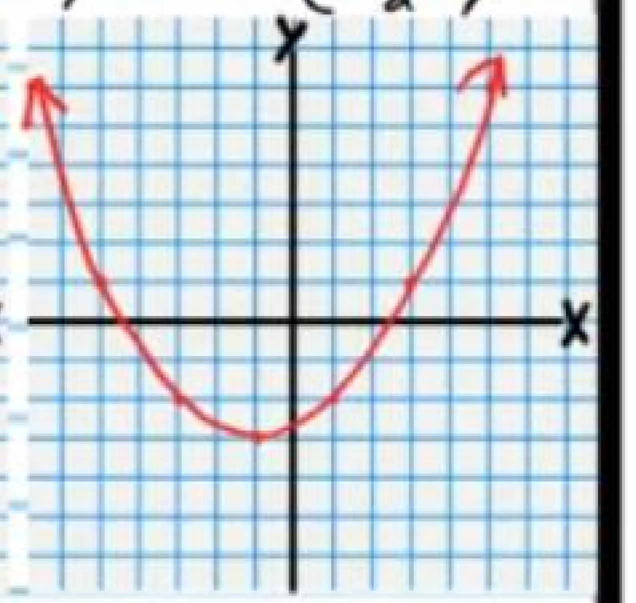
⑥ $y+3=\left(\frac{-x-1}{2}\right)^2$

Parent Relation	Transformation #1	Transformation #2	Final Transformation
	Dilation	Translation	Reflection
$y=x^2$	$y=\left(\frac{x}{2}\right)^2$	$y+3=\left(\frac{x-1}{2}\right)^2$	$y+3=\left(\frac{-x-1}{2}\right)^2$



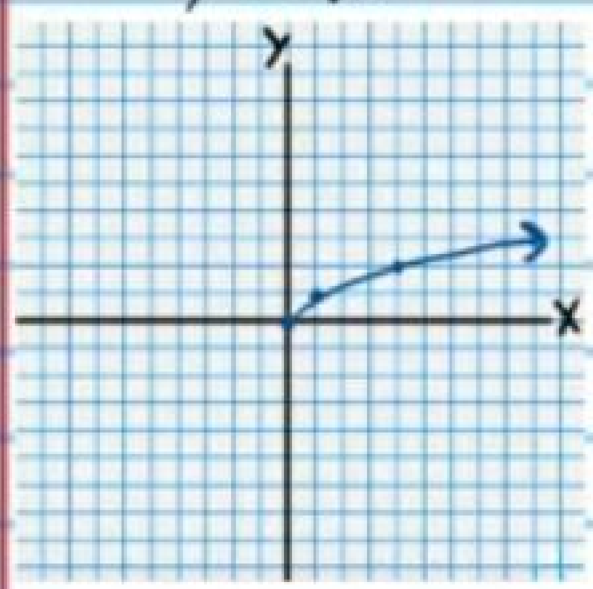


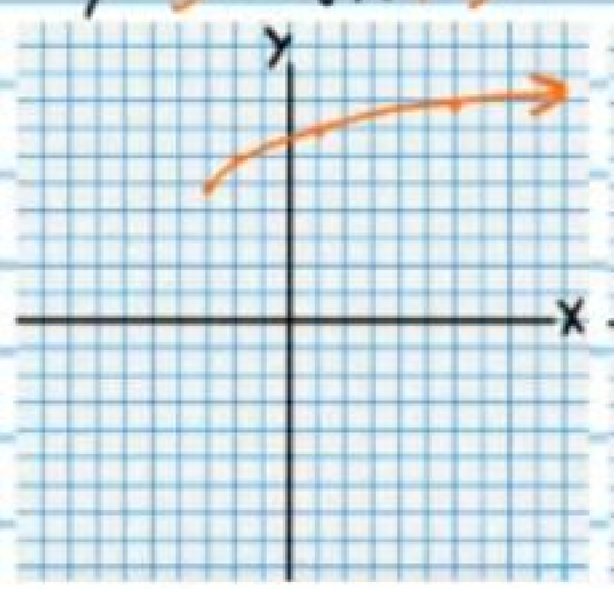


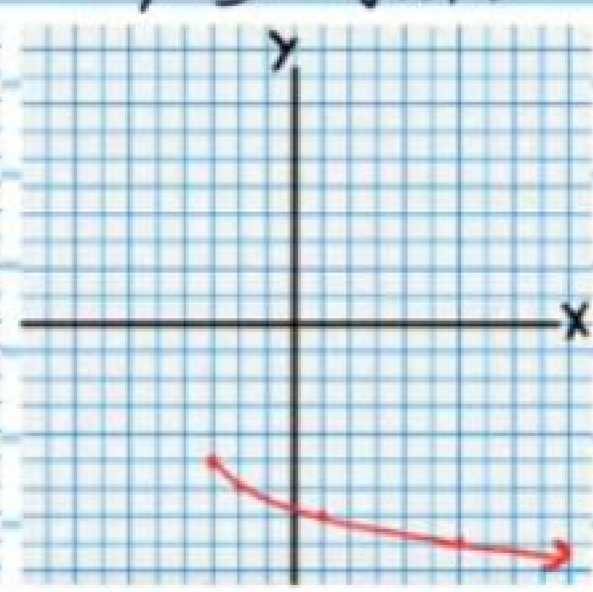


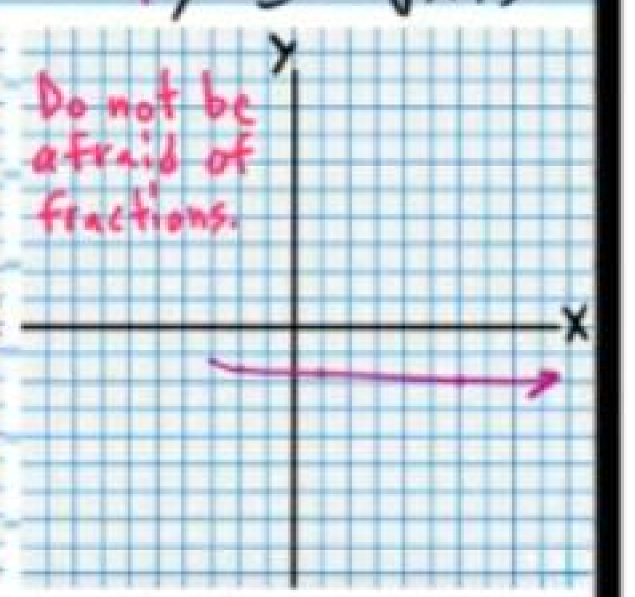
⑦ $-4y-5=\sqrt{x+3}$

Parent Relation	Transformation #1	Transformation #2	Final Transformation
	Translation	Reflection	Dilation
$y=\sqrt{x}$	$y-5=\sqrt{x+3}$	$-y-5=\sqrt{x+3}$	$-4y-5=\sqrt{x+3}$





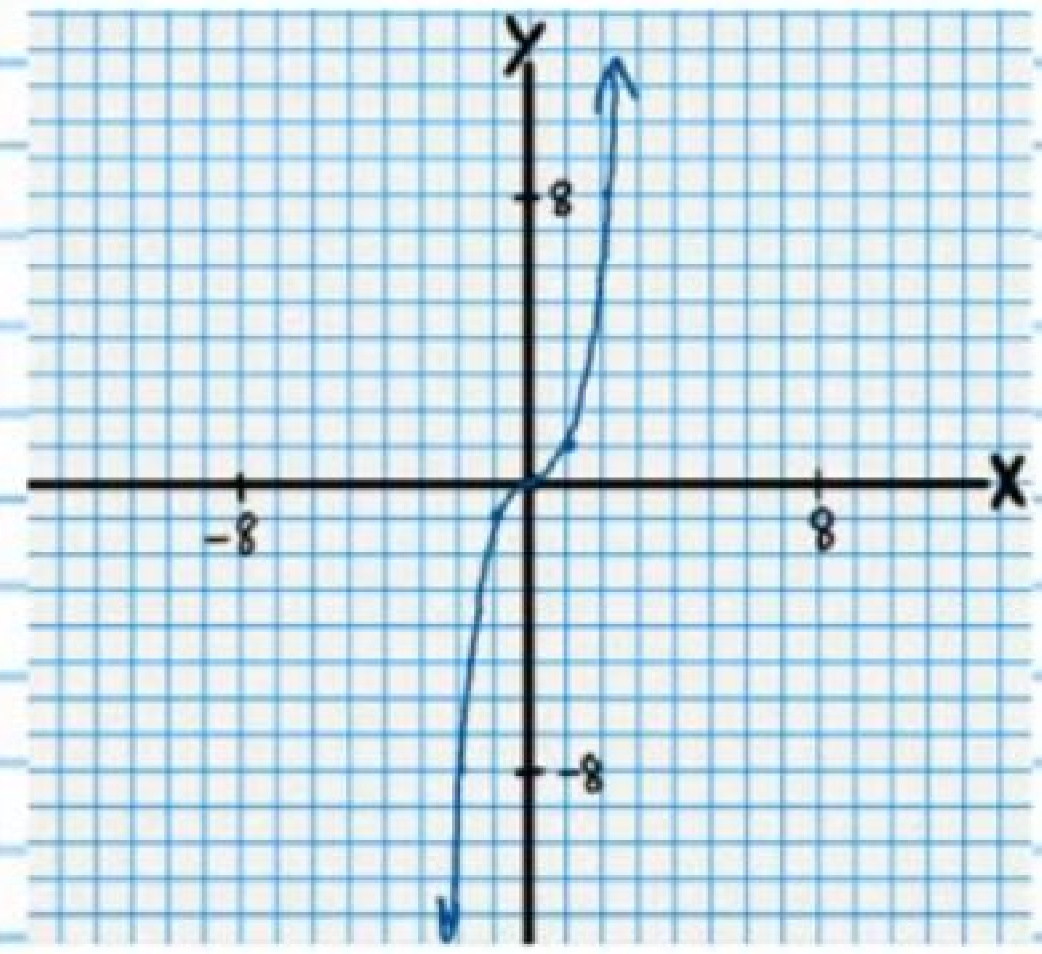




⑧ $\frac{y-2}{3}=\left[-(x-4)\right]^3$

Parent Relation

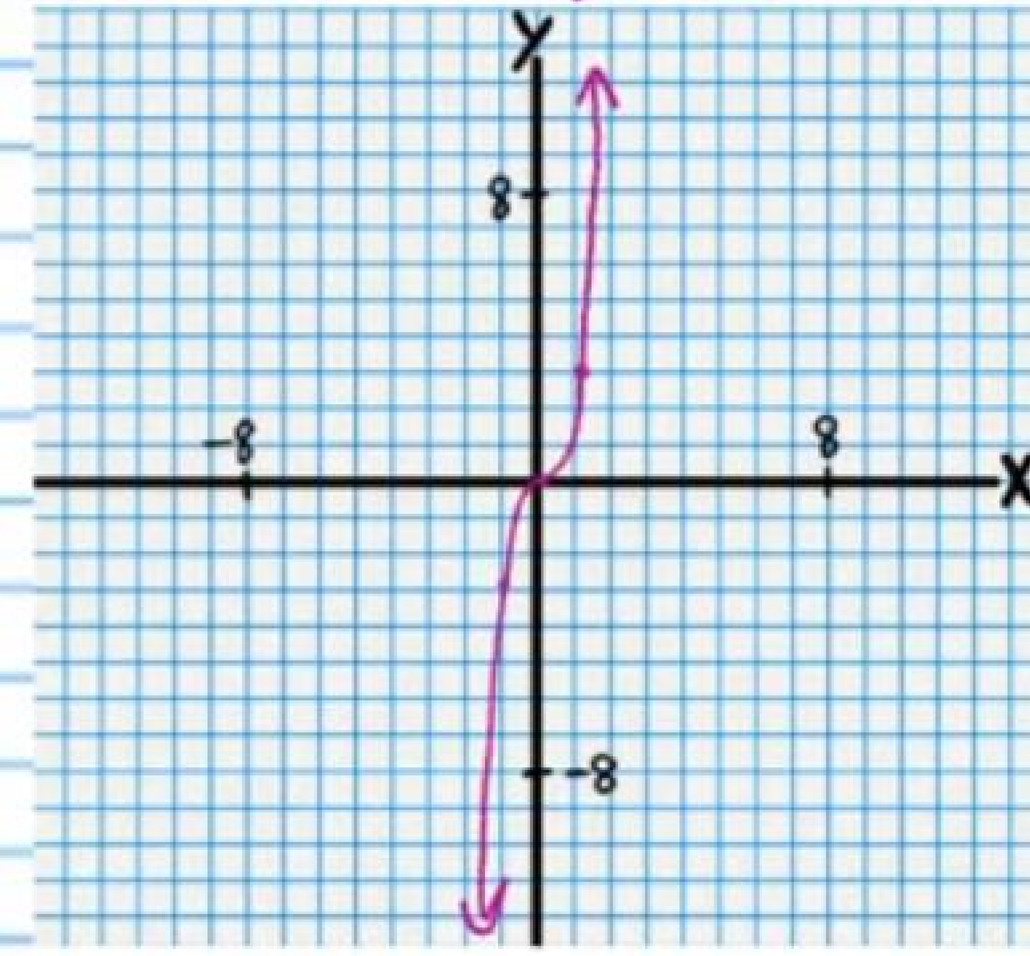
$y=x^3$



Transformation #1

Dilation

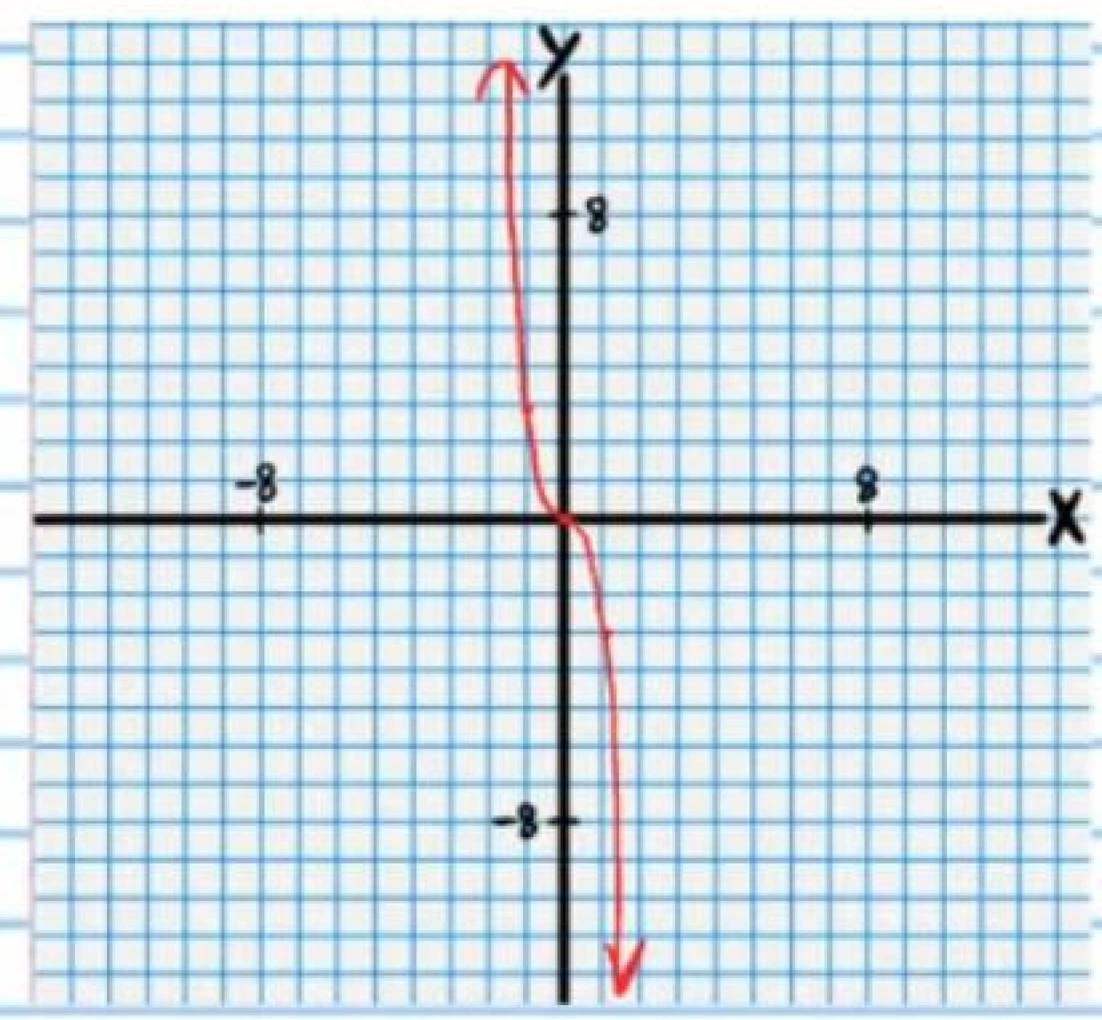
$\frac{y}{3}=x^3$



Transformation #2

Reflection

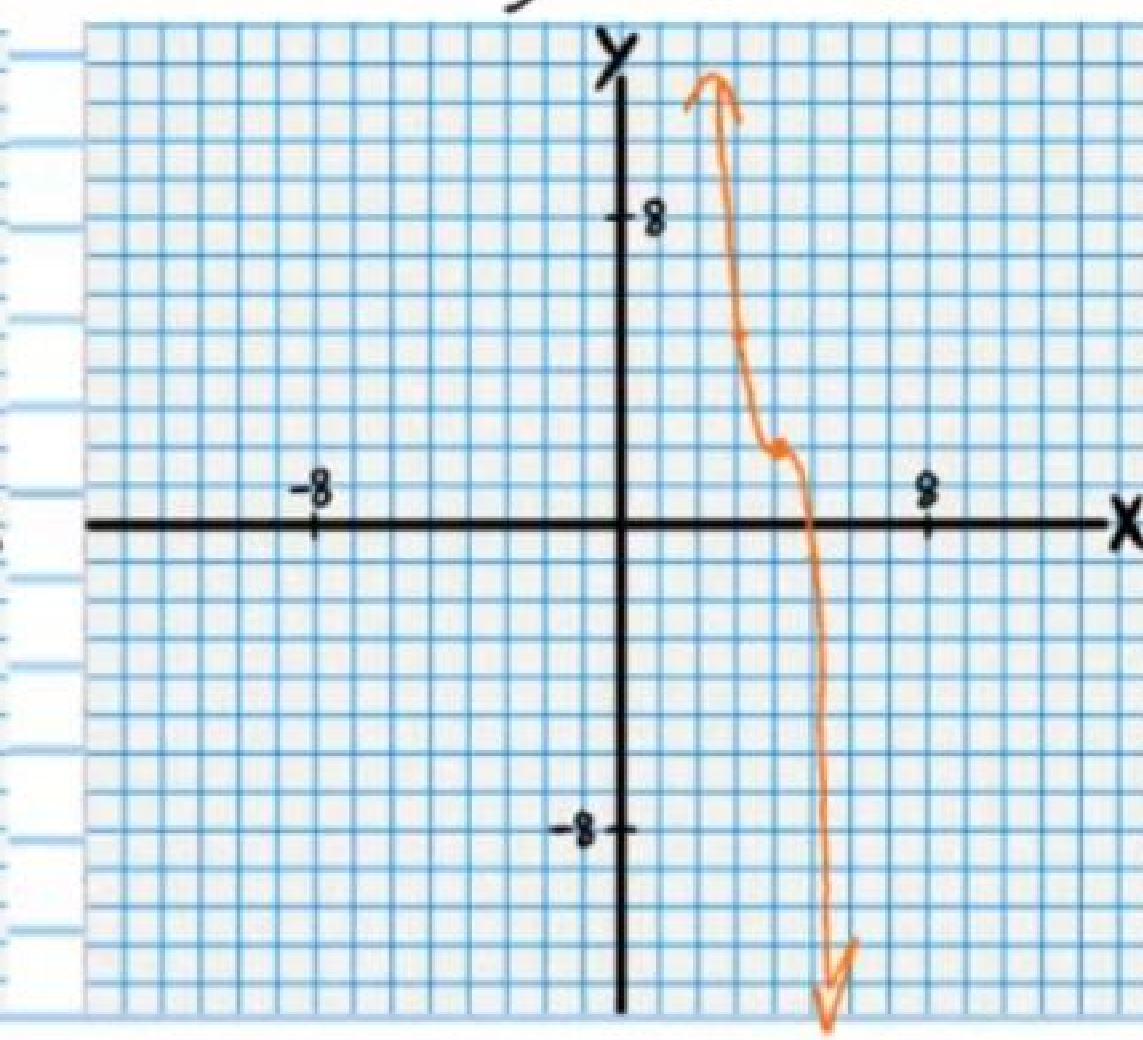
$\frac{y}{3}=\left[-x\right]^3$



Final Transformation

Translation

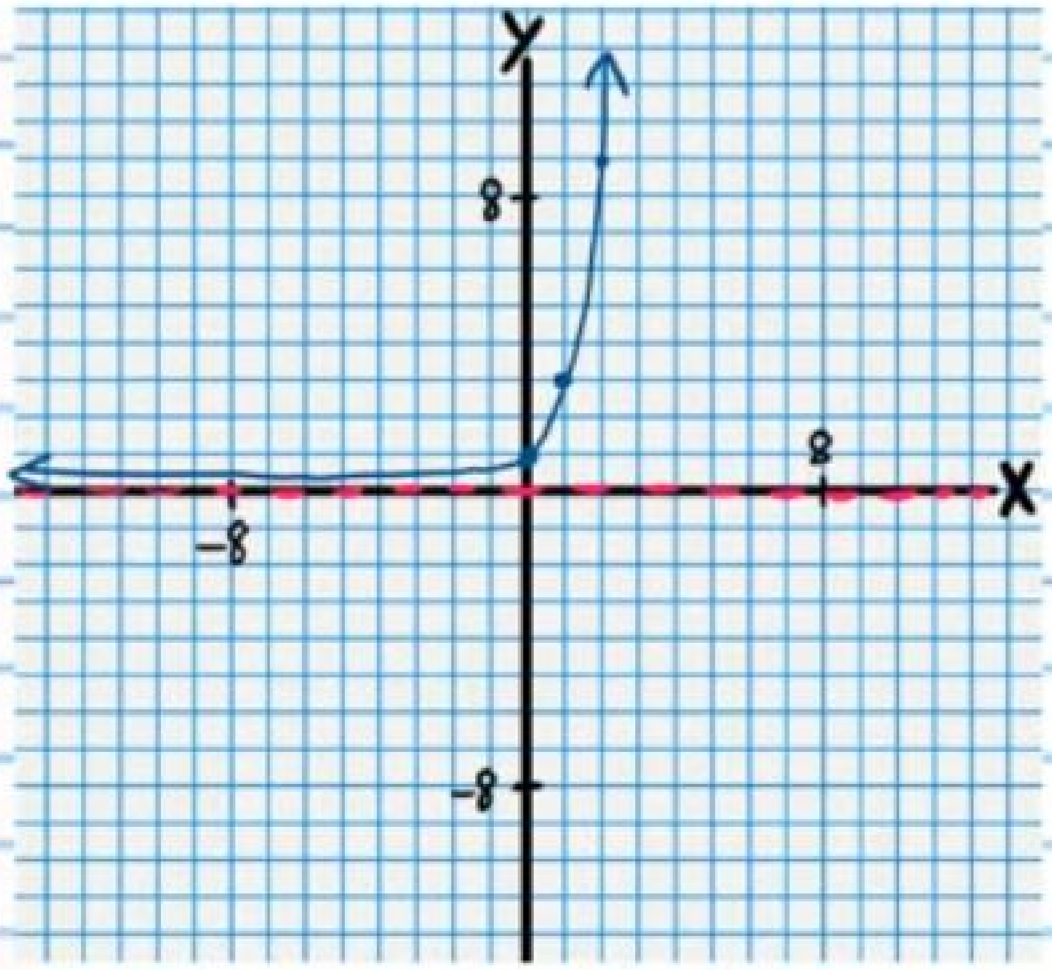
$\frac{y-2}{3}=\left[-(x-4)\right]^3$



⑨ $-(y-2) = 3^{\frac{x}{4}+2}$

Parent Relation

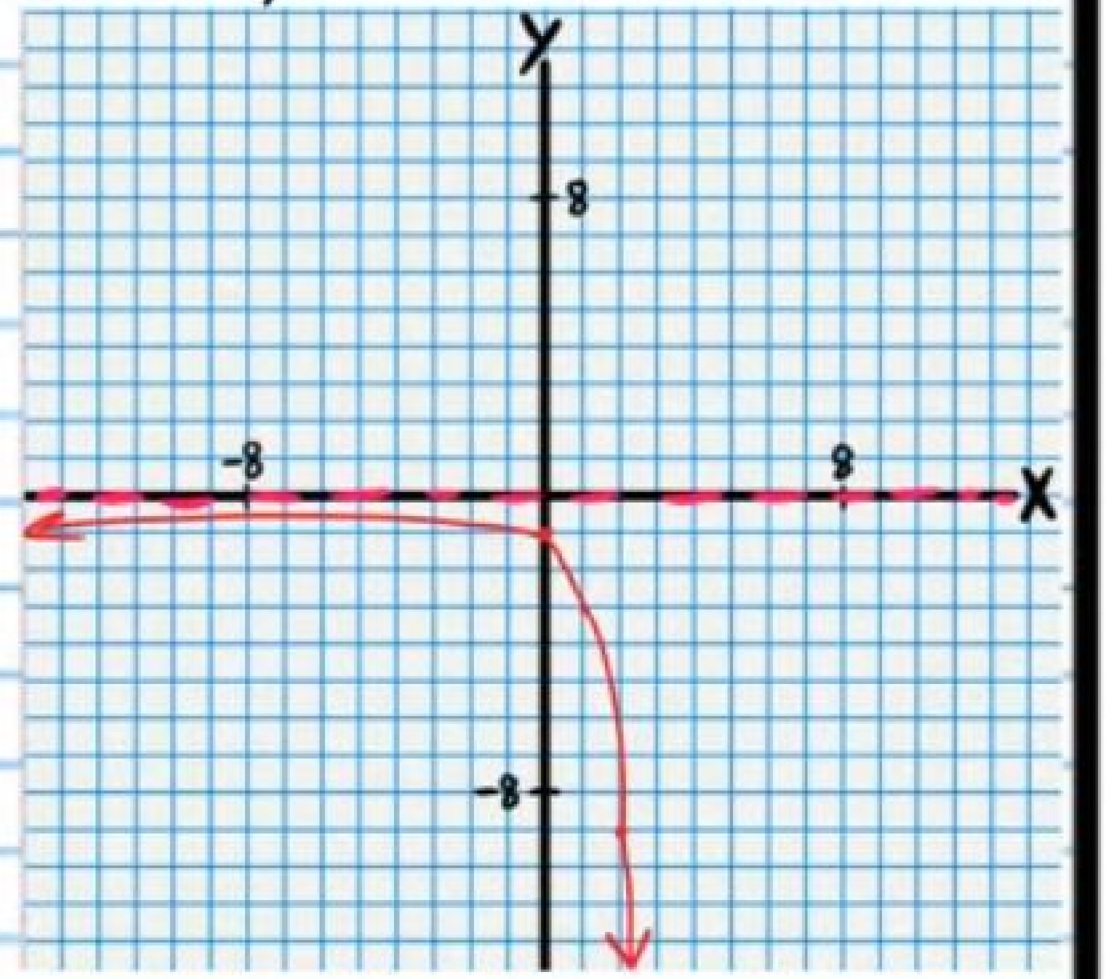
$y = 3^x$



Transformation #1

Reflection

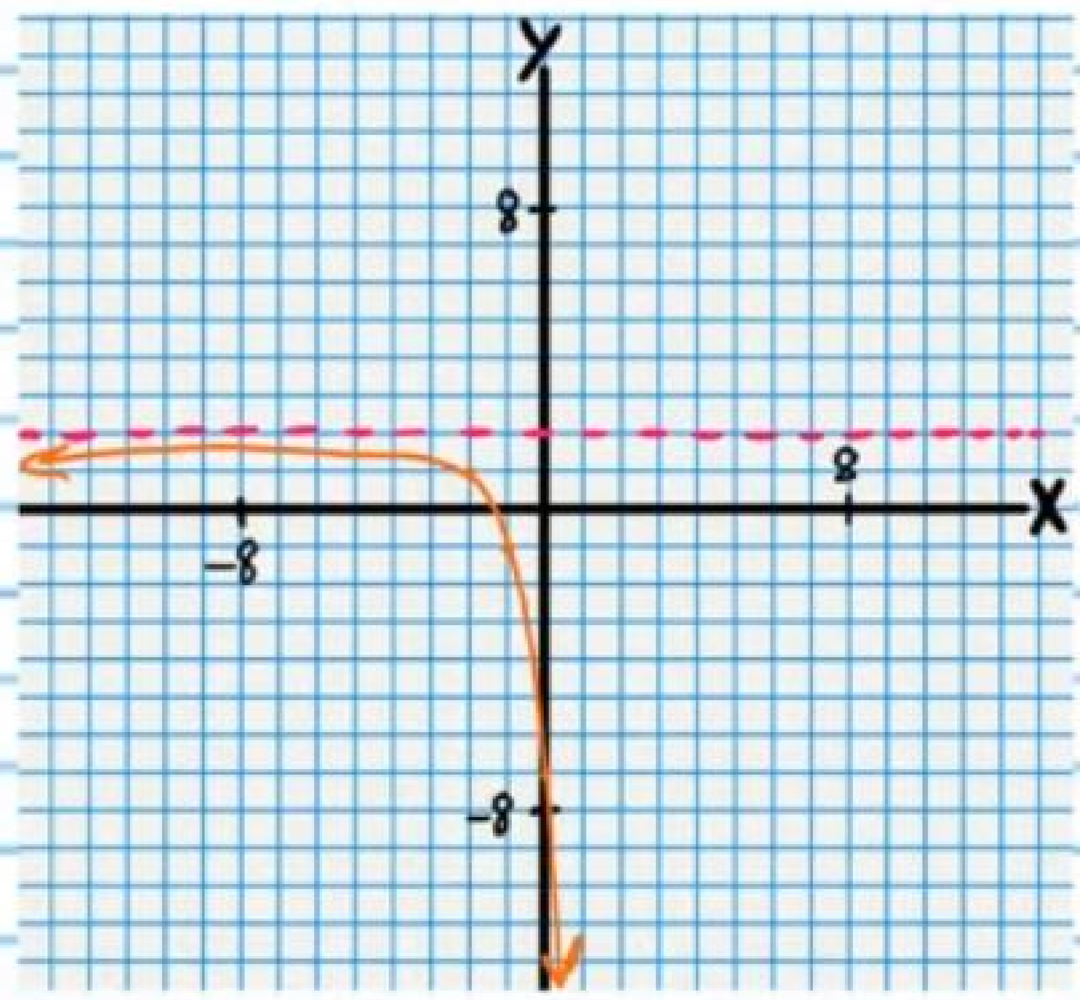
$-y = 3^x$



Transformation #2

Translation

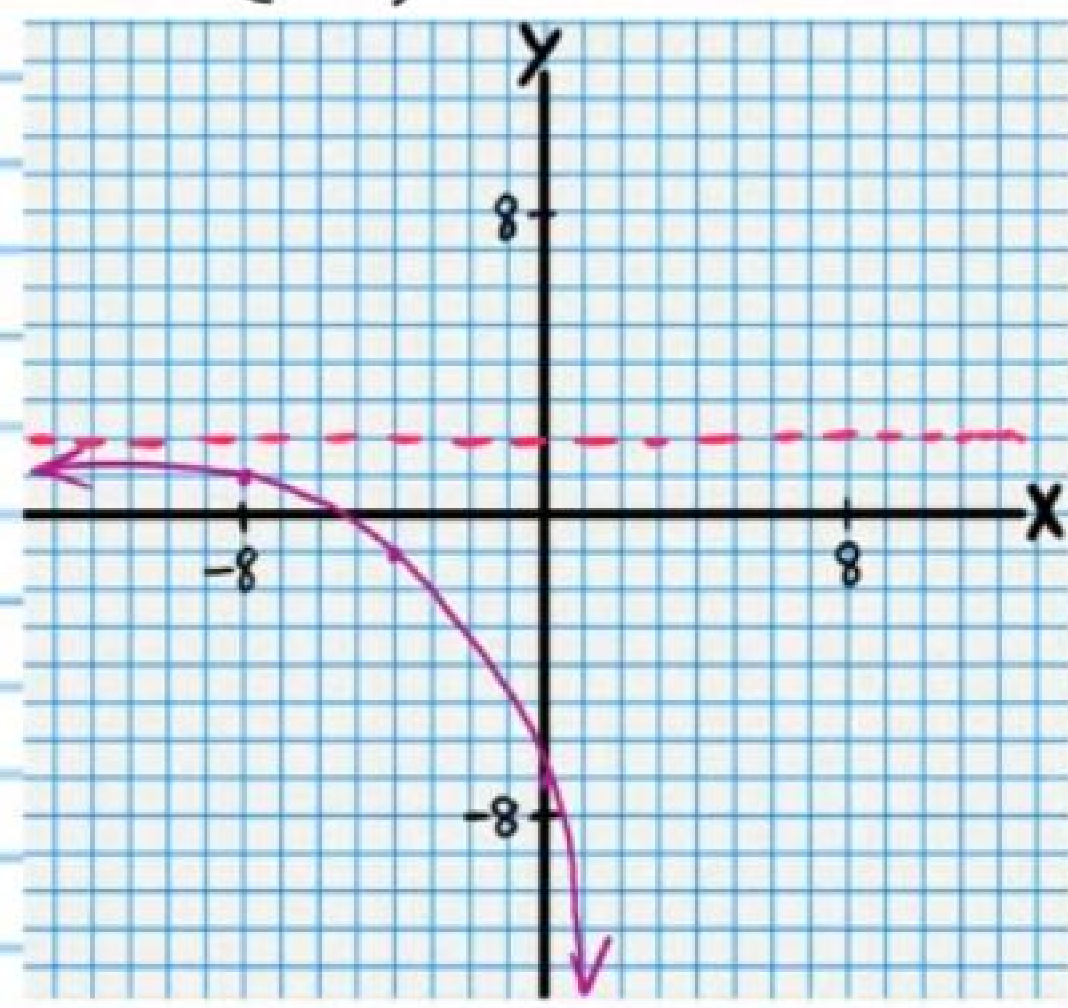
$-(y-2) = 3^{x+2}$



Final Transformation

Dilation

$-(y-2) = 3^{\frac{x}{4}+2}$

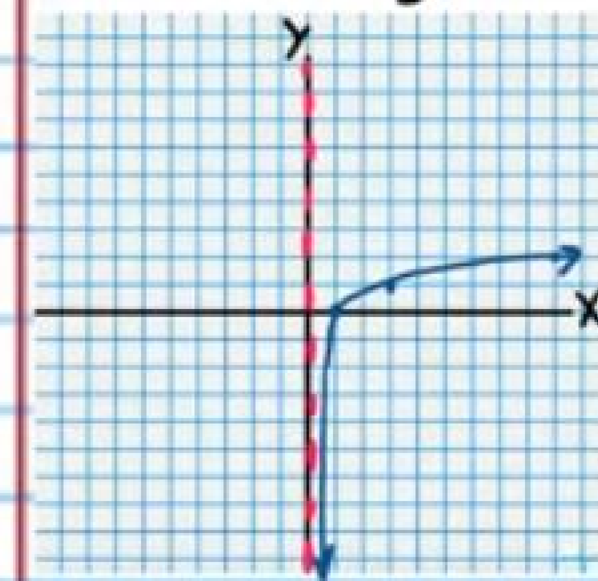


Logarithm Property III	Logarithm Property II	Logarithm Property IX
$\log_b b = 1$	$\log_b b^x = x$	$\log_b 1 = 0$

⑩ $y+3 = \log_3(-3x-6)$

Parent Relation

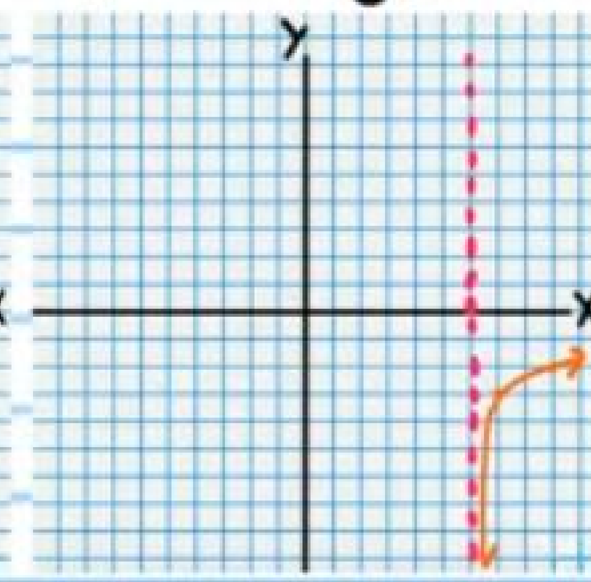
$y = \log_3 x$



Transformation #1

Translation

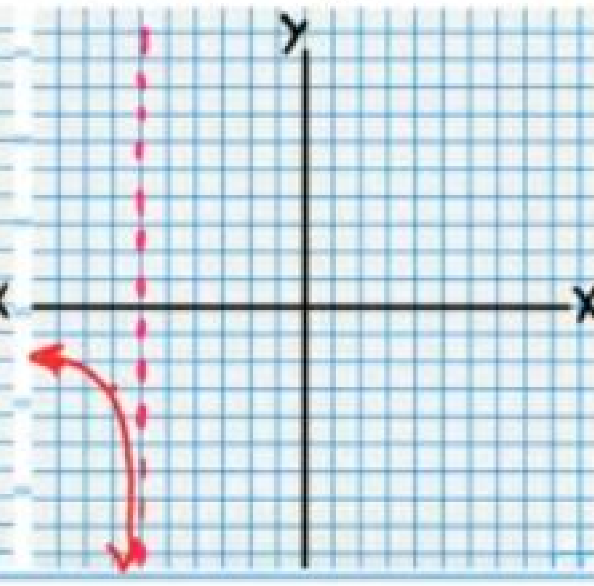
$y+3 = \log_3(x-6)$



Transformation #2

Reflection

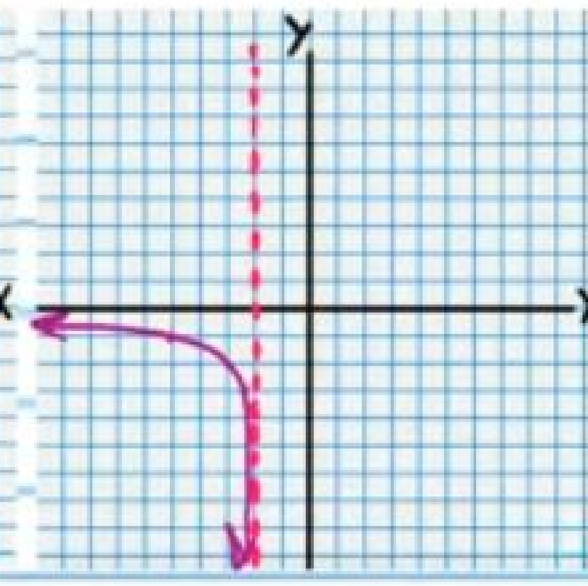
$y+3 = \log_3(-x-6)$



Final Transformation

Dilation

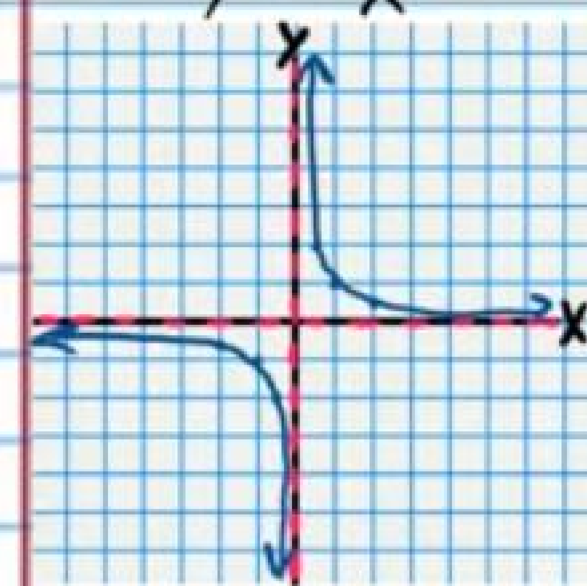
$y+3 = \log_3(-3x-6)$



⑪ $-\frac{y-1}{2} = \frac{1}{x-3}$

Parent Relation

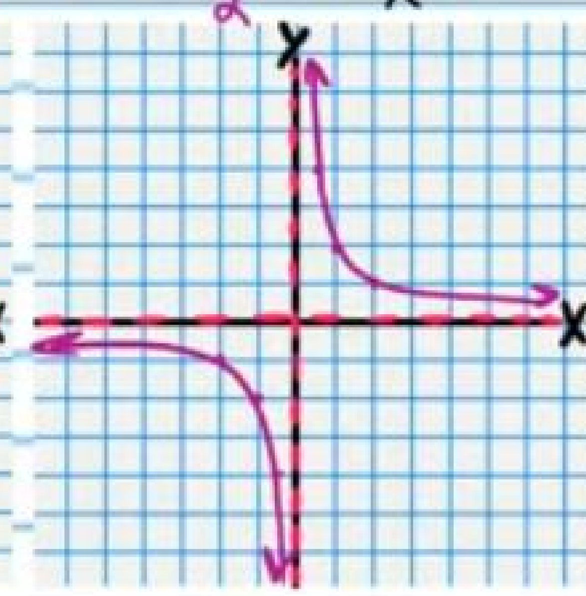
$y = \frac{1}{x}$



Transformation #1

Dilation

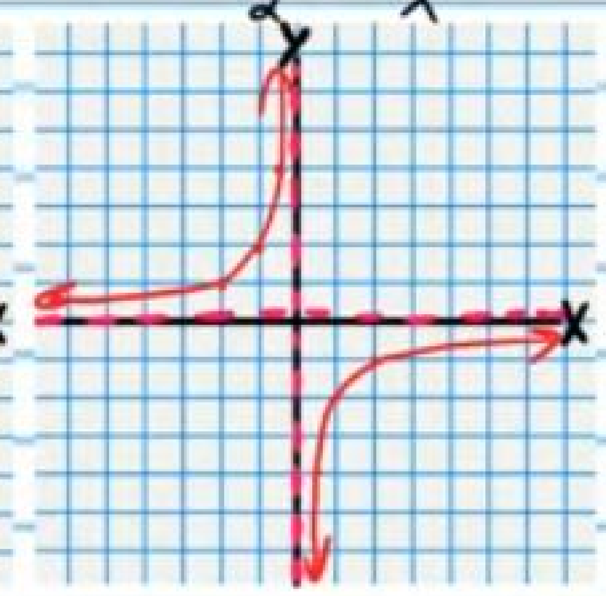
$\frac{y}{2} = \frac{1}{x}$



Transformation #2

Reflection

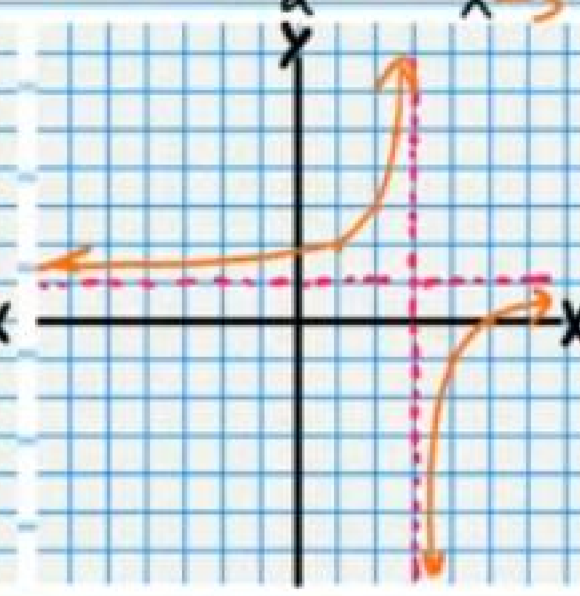
$\frac{-y}{2} = \frac{1}{x}$



Final Transformation

Translation

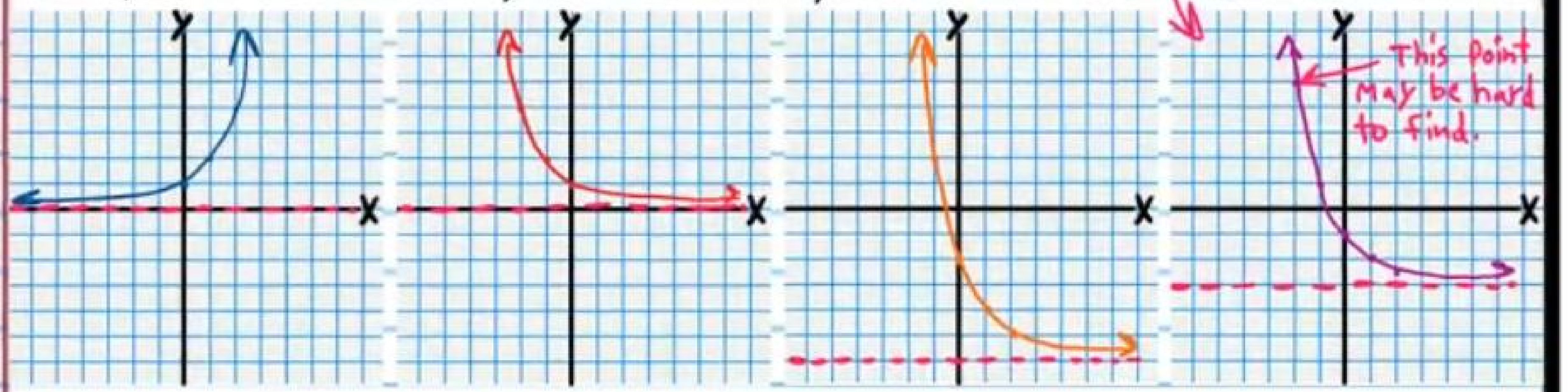
$-\frac{y-1}{2} = \frac{1}{x-3}$



You will need to move the asymptote in the last step.

⑫ $2y+6=2^{-(x-2)}$

Parent Relation	Transformation #1	Transformation #2	Final Transformation
	Reflection	Translation	Dilation
$y=2^{x}$	$y=2^{-x}$	$y+6=2^{-(x-2)}$	$2y+6=2^{-(x-2)}$



Logarithm Property III

$\log_b b=1$

Logarithm Property II

$\log_b b^{x}=x$

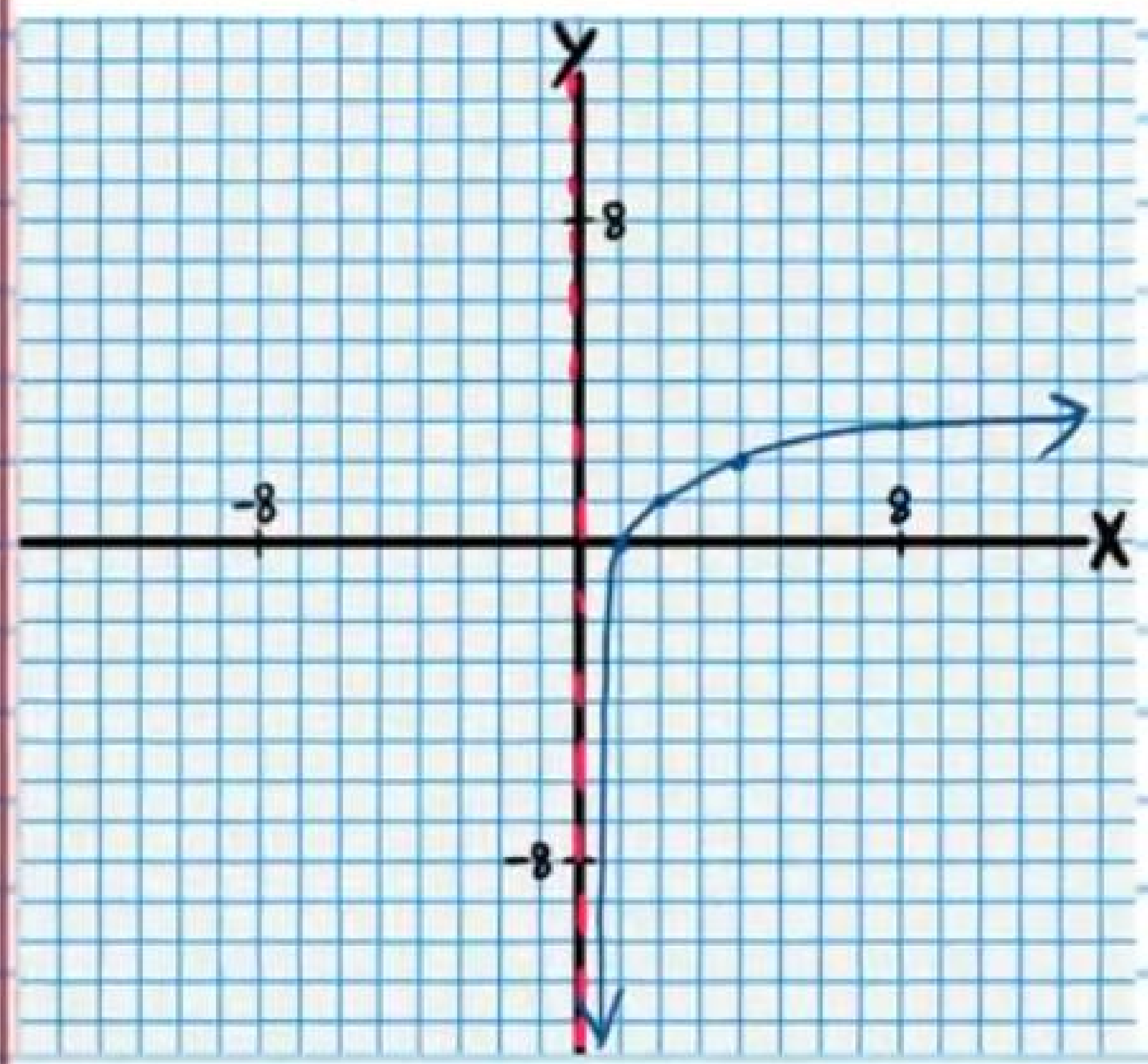
Logarithm Property IX

$\log_b 1=0$

⑬ $-y+4=\log_2\left(\frac{x}{3}+1\right)$

Parent Relation

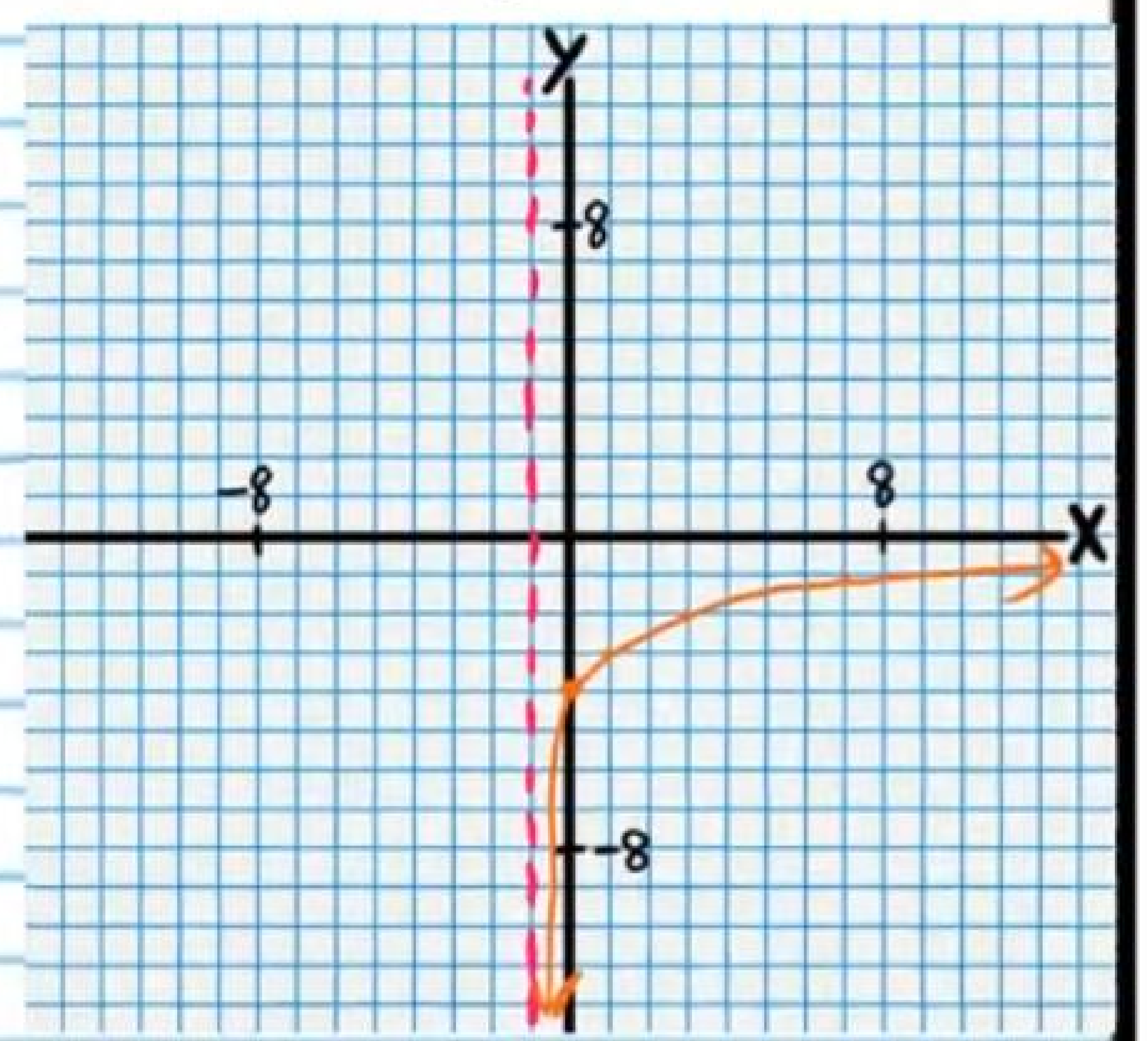
$y=\log_2 x$



Transformation #1

Translation

$y+4=\log_2(x+1)$

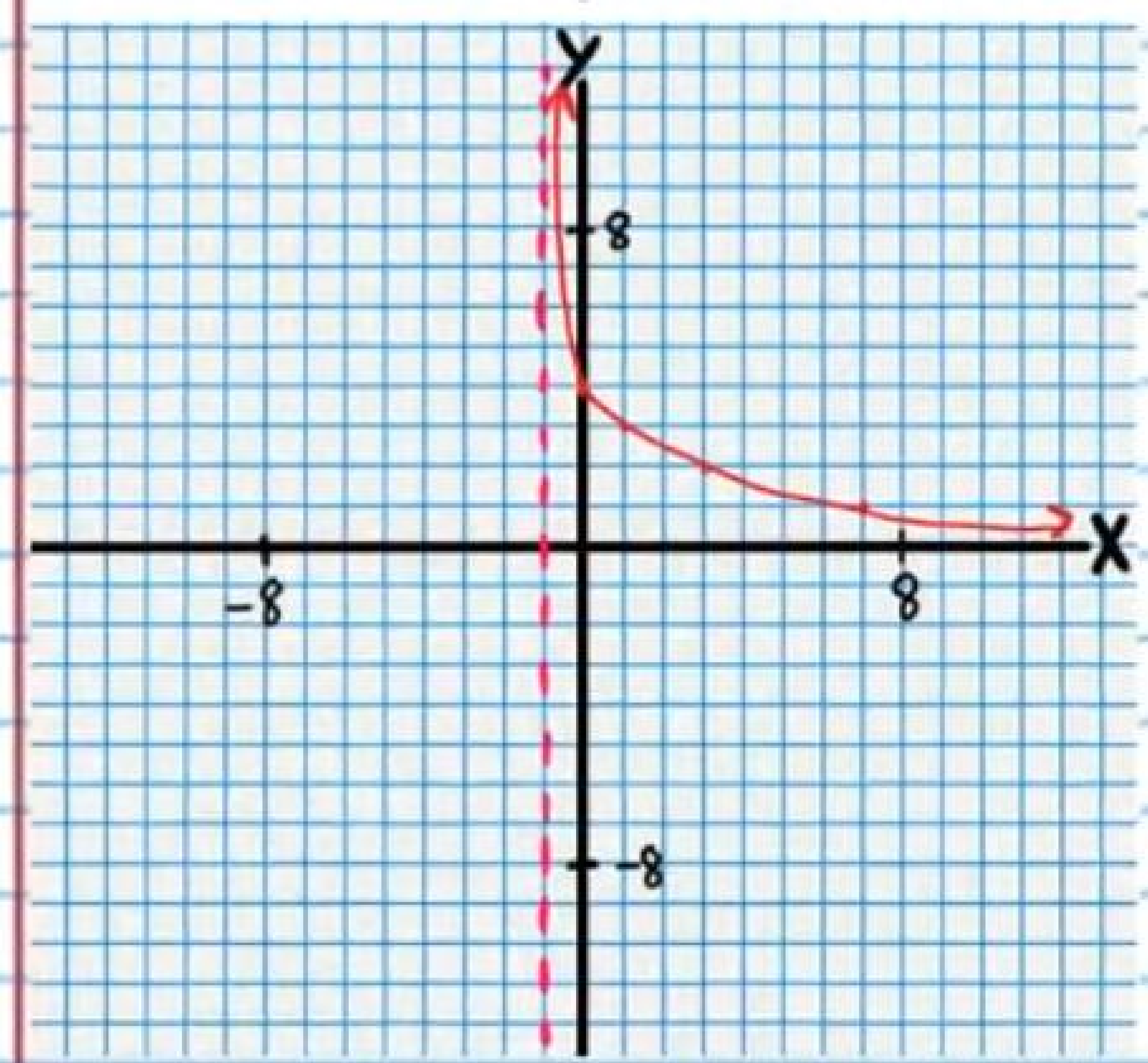


You will need to move the asymptote in these steps.

Transformation #2

Reflection

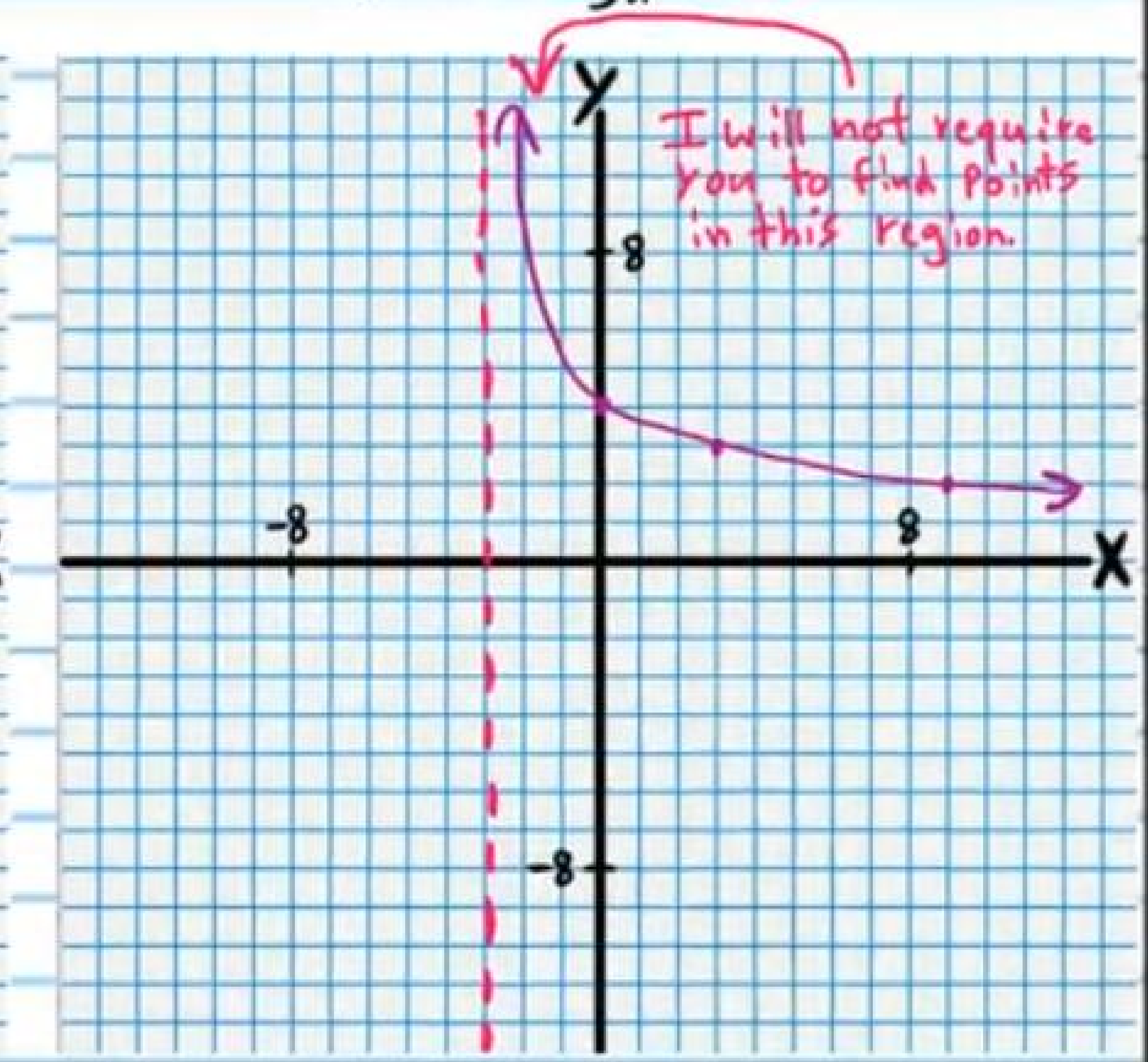
$-y+4=\log_2(x+1)$



Final Transformation

Dilation

$-y+4=\log_2\left(\frac{x}{3}+1\right)$



⑭ $2(y-1)=\frac{1}{-(x+2)}$

Parent Relation	Transformation #1	Transformation #2	Final Transformation
	Reflection	Dilation	Translation
$y=\frac{1}{x}$	$y=\frac{1}{-x}$	$2y=\frac{1}{-x}$	$2(y-1)=\frac{1}{-(x+2)}$